MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE LHP

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

* 1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2 APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.24 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be LHP.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/75-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

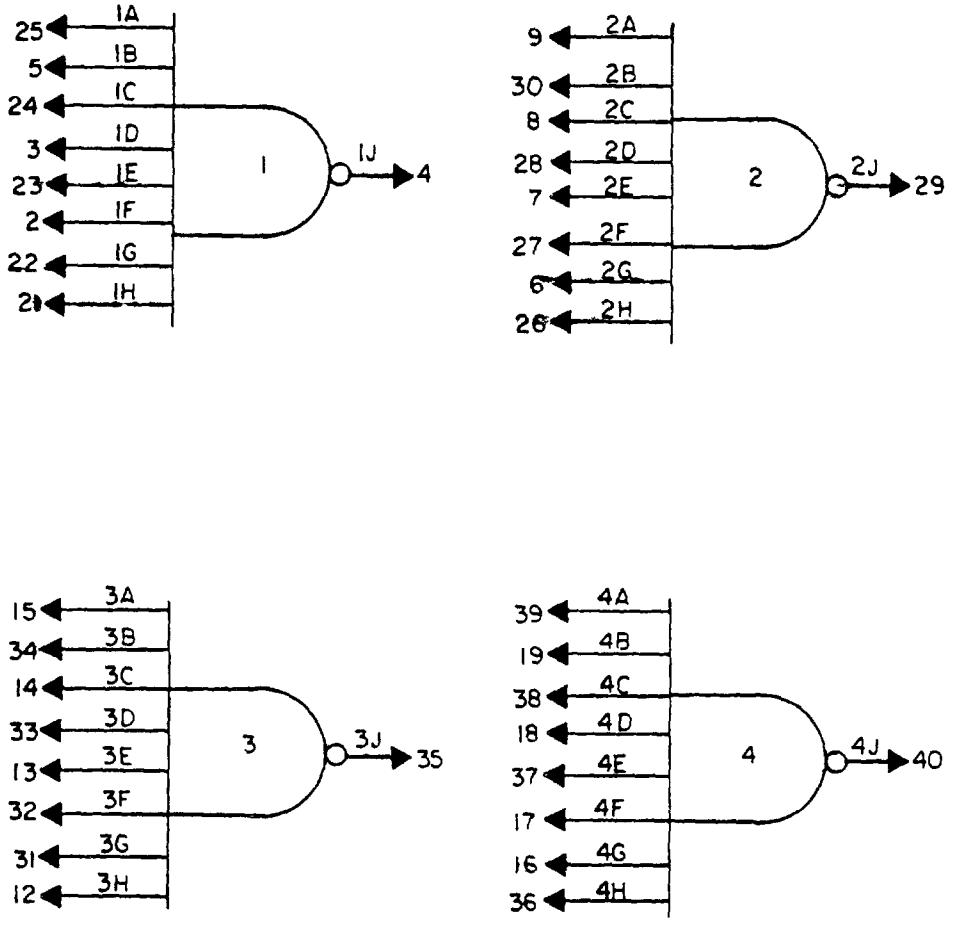
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. 1A-1H, 2A-2H, 3A-3H, 4A-4H = Data inputs.
2. 1J-4J = Data outputs.
3. Pins 11 and 20 are not used.

* FIGURE 1. LHP module function diagram, NAND gate.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A3	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B4	V_{OUT} (low loaded)	IV	S	0.01	1.00	0.01	1.00	0.01	1.00	0.01	1.00	V
C1-C32	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
D1-D4	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
E1-E4	Output leakage current	VII	X	0	1	0	1	0	1	0	1	mA
F1-F32	Input clamp diode voltage	VIII	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G4	Output voltage, low level	IX	X	0	500	0	500	0	500	0	500	mV
H1-H32	Output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1-J32	Input current, high level	XI	X	0	50	0	50	0	50	0	50	µA
K1-K32	Input current, low level	XII	X	-0.4	-2.2	0.4	-2.2	-0.4	-2.2	-0.4	-2.2	mA
L1	Power supply current, low level	XIII	S	5	40	5	40	5	40	5	40	mA
L2	Power supply current, high level	XIII	S	2	20	2	20	2	20	2	20	mA
M1-M4	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	2	15	2	15	2	18	2	18	ns
M5-M8	Transition time (t_{TLH}), C_L = 150 pF	XIV	Q	6	27	6	27	6	30	6	30	ns
N1-N4	Transition time (t_{THL}), C_L = 50 pF	XIV	S	2	6	2	6	2	8	2	8	ns
N5-N8	Transition time (t_{THL}), C_L = 150 pF	XIV	Q	3	11	3	11	3	13	3	13	ns
P1,P9, P17,P25	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	S	2	8	2	8	2	9	2	9	ns
P2-P8,P10-P16,P18-P24, P26-P32	Propagation delay time (t_{PLH}), C_L = 50 pF	XIV	Q	2	8	2	8	2	9	2	9	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P33-P64	Propagation delay time (t_{PLH}), C_L = 150 pF	XIV	Q	3	12	3	12	3.0	12.5	3.0	12.5	ns
Q1,Q9, Q17,Q25	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	S	2	9	2	9	2	10	2	10	ns
Q2-Q8,Q10-Q16,Q18-Q24, Q26-Q32	Propagation delay time (t_{PHL}), C_L = 50 pF	XIV	Q	2	9	2	9	2	10	2	10	ns
Q33-Q64	Propagation delay time (t_{PHL}), C_L = 150 pF	XIV	Q	3	13	3	13	3	14	3	14	ns
R1-R4	Output crosstalk, low level	XV	Q	0	500	0	500	0	500	0	500	mV
S1-S4	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	mV
T	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XVII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
V1-V32	Input capacitance	XVIII	Q	5	18	5	18	5	18	5	18	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through V shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/75-1 superseded part number M28787/75. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation (V_{CC} = 5.5 V)	
Outputs low	220 mW (outputs not loaded)
Outputs high	110 mW (outputs not loaded)
Supply voltage	5 V dc ±10 percent
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	2 V minimum, V_{CC} maximum

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Input current		
Low level	(V_{IL} = 0 V)	-2.2 mA maximum
	(V_{IL} = 0.5 V)	-2 mA maximum
High level	(V_{IH} = 2.5 V)	50 µA maximum
	(V_{IH} = 5.5 V)	1 mA maximum
Output voltage		
Low level	(I_{OL} = 20 mA)	0.5 V maximum
High level	(I_{OH} = -1 mA)	2.5 V minimum
Output drive current maximum capability		
Low level	(V_{OL} = 0.5 V)	20 mA
High level	(V_{OH} = 2.5 V)	-1 mA

* 6.5 Module function. This module contains four independent 8-input NAND gates.

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

TABLE II. Input to output relationship. 1/ 2/

Inputs								Output
A	B	C	D	E	F	G	H	J
H	H	H	H	H	H	H	H	L
L	H	H	H	H	H	H	H	H
H	L	H	H	H	H	H	H	H
H	H	L	H	H	H	H	H	H
H	H	H	L	H	H	H	H	H
H	H	H	H	L	H	H	H	H
H	H	H	H	H	L	H	H	H
H	H	H	H	H	H	L	H	H
H	H	H	H	H	H	H	L	H
L	L	L	L	L	L	L	L	H

1/ H = High level.
2/ L = Low level.

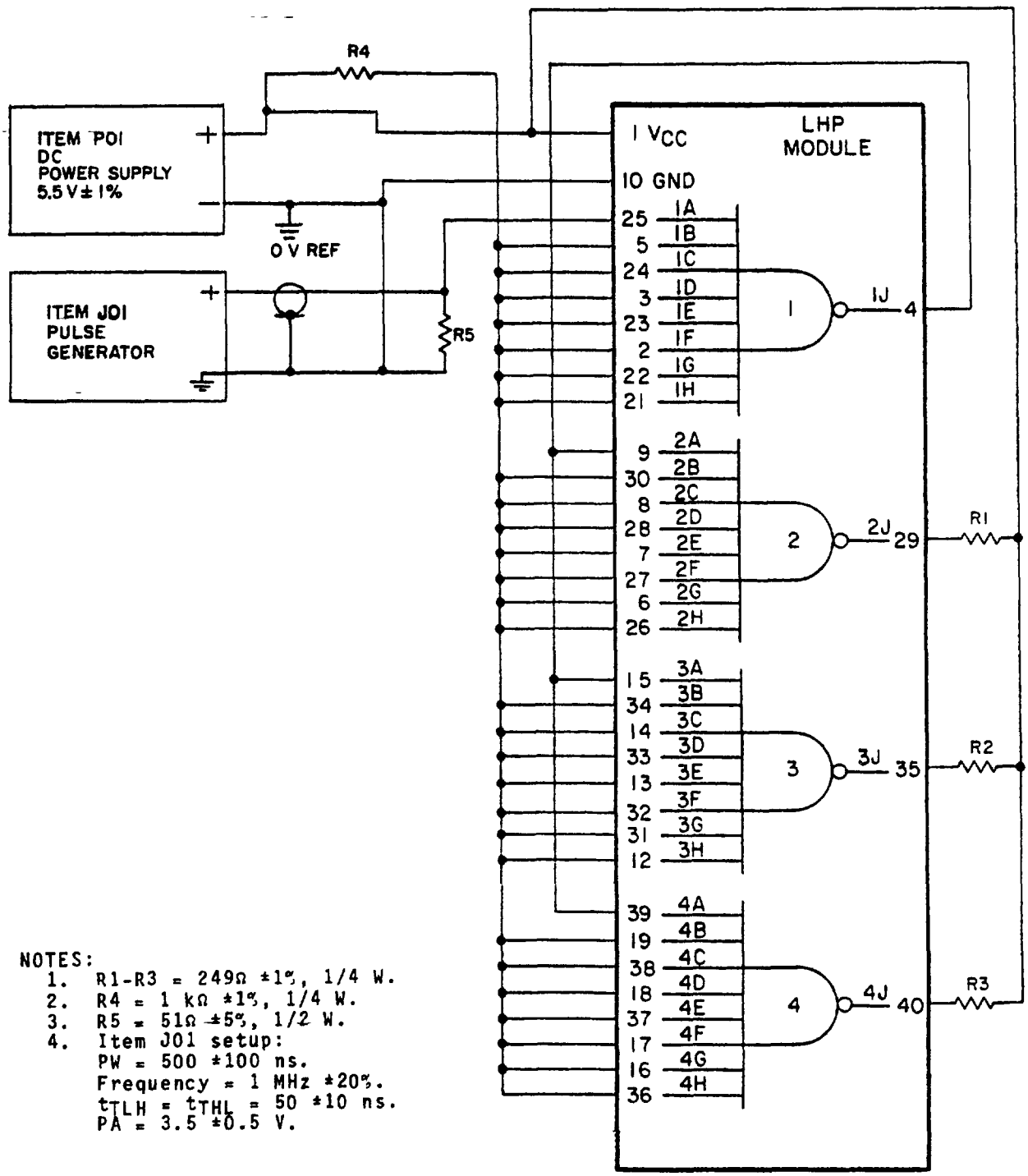
6.6 Changes from previous issue The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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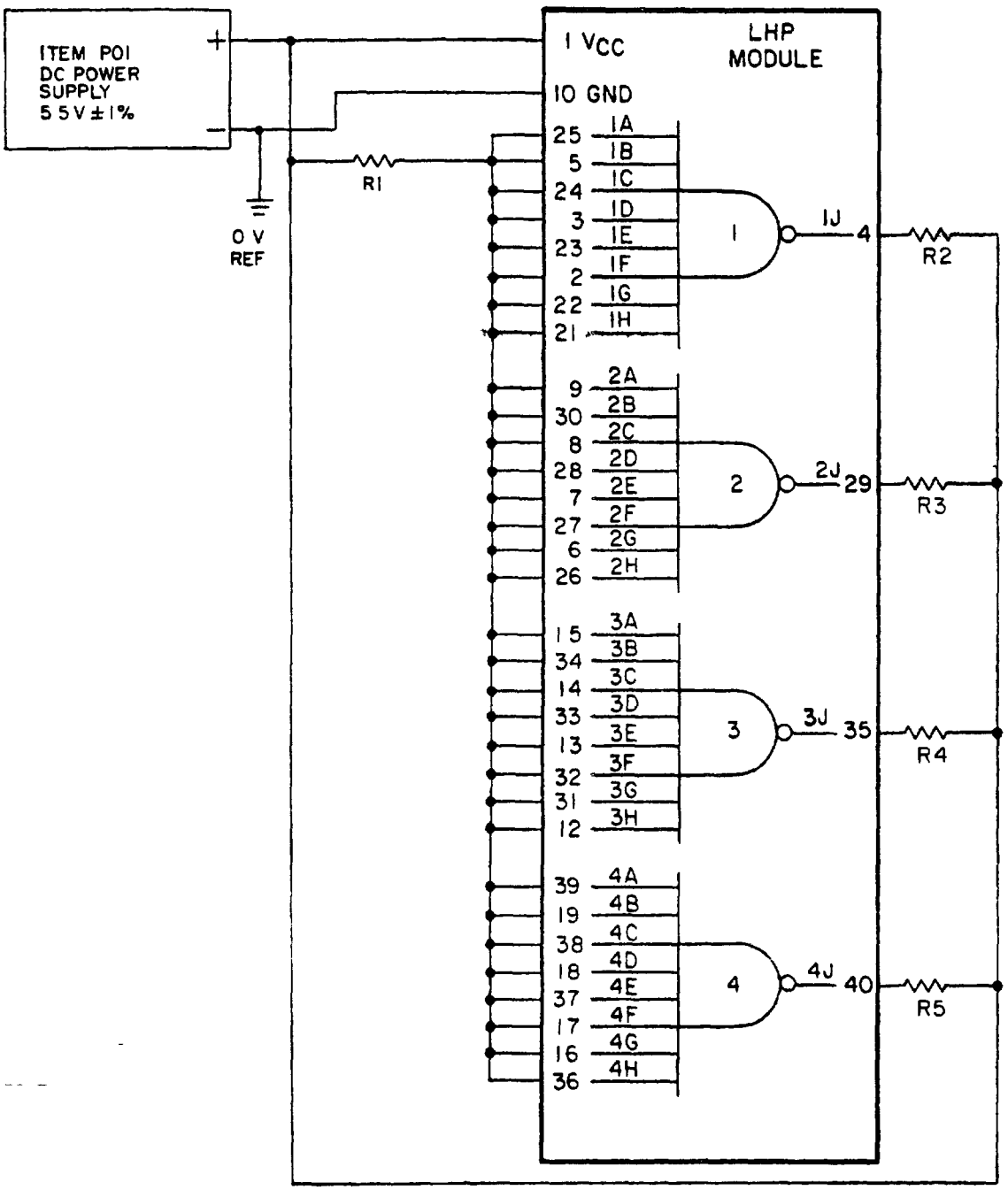
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NOTES:
1. R1-R3 = 249Ω ±1%, 1/4 W.
2. R4 = 1 kΩ ±1%, 1/4 W.
3. R5 = 51Ω ±5%, 1/2 W.
4. Item J01 setup:
   PW = 500 ±100 ns.
   Frequency = 1 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.
   PA = 3.5 ±0.5 V.

* FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 1 kΩ ±5%, 1/4 W.
2. R2-R5 = 249Ω ±1%, 1/4 W.

FIGURE 3 Component temperature test circuit.

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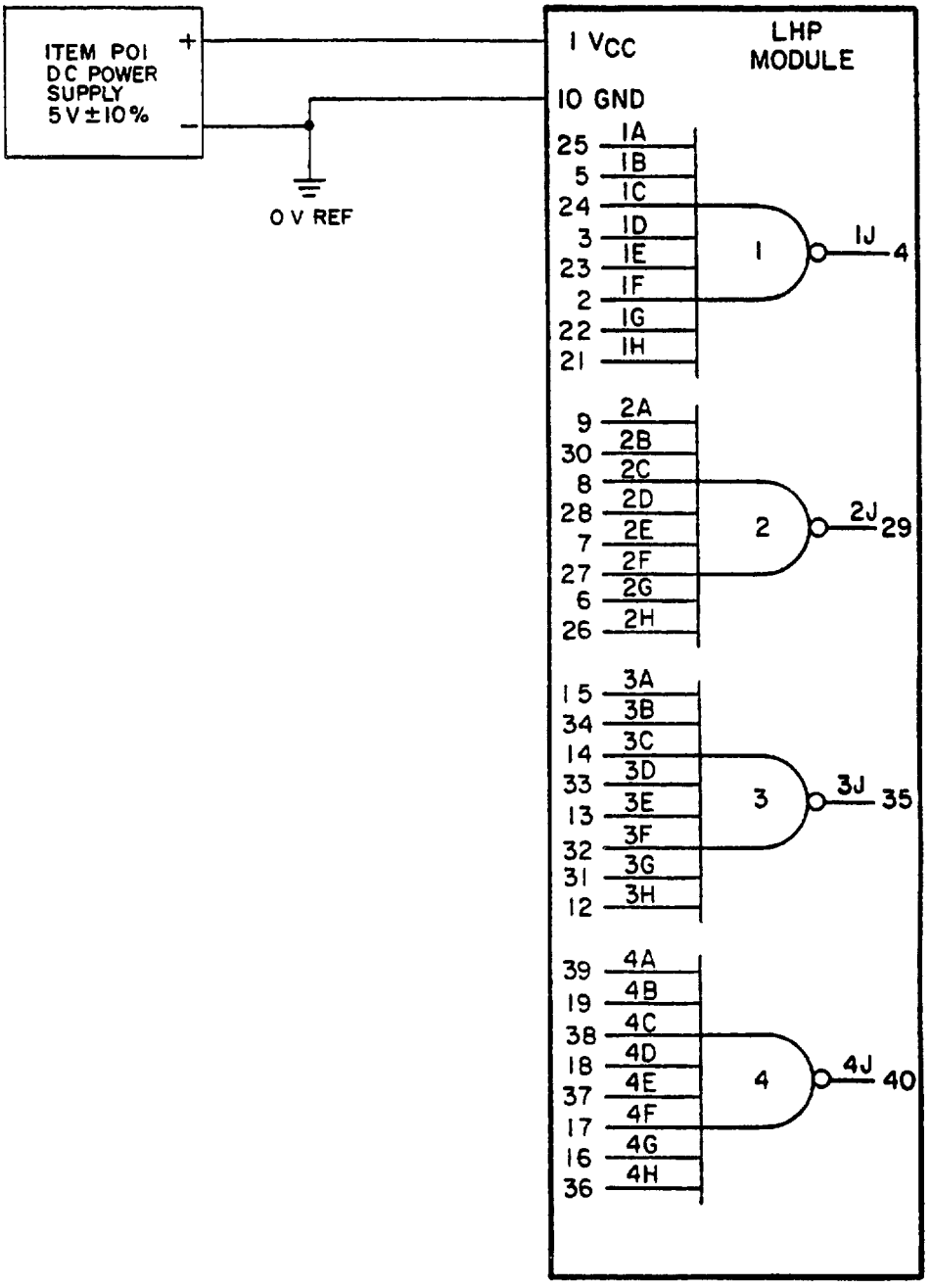


FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C plus 11 and 20	Frame	10	∞	MΩ	5
A2	"	C	11 and 20	"	"	"	"
A3	"	11	20	"	"	"	"

TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1J V_{OUT} (low loaded)	4	0.01	1.00	V	6
B2	2J "	29	"	"	"	"
B3	3J "	35	"	"	"	"
B4	4J "	40	"	"	"	"

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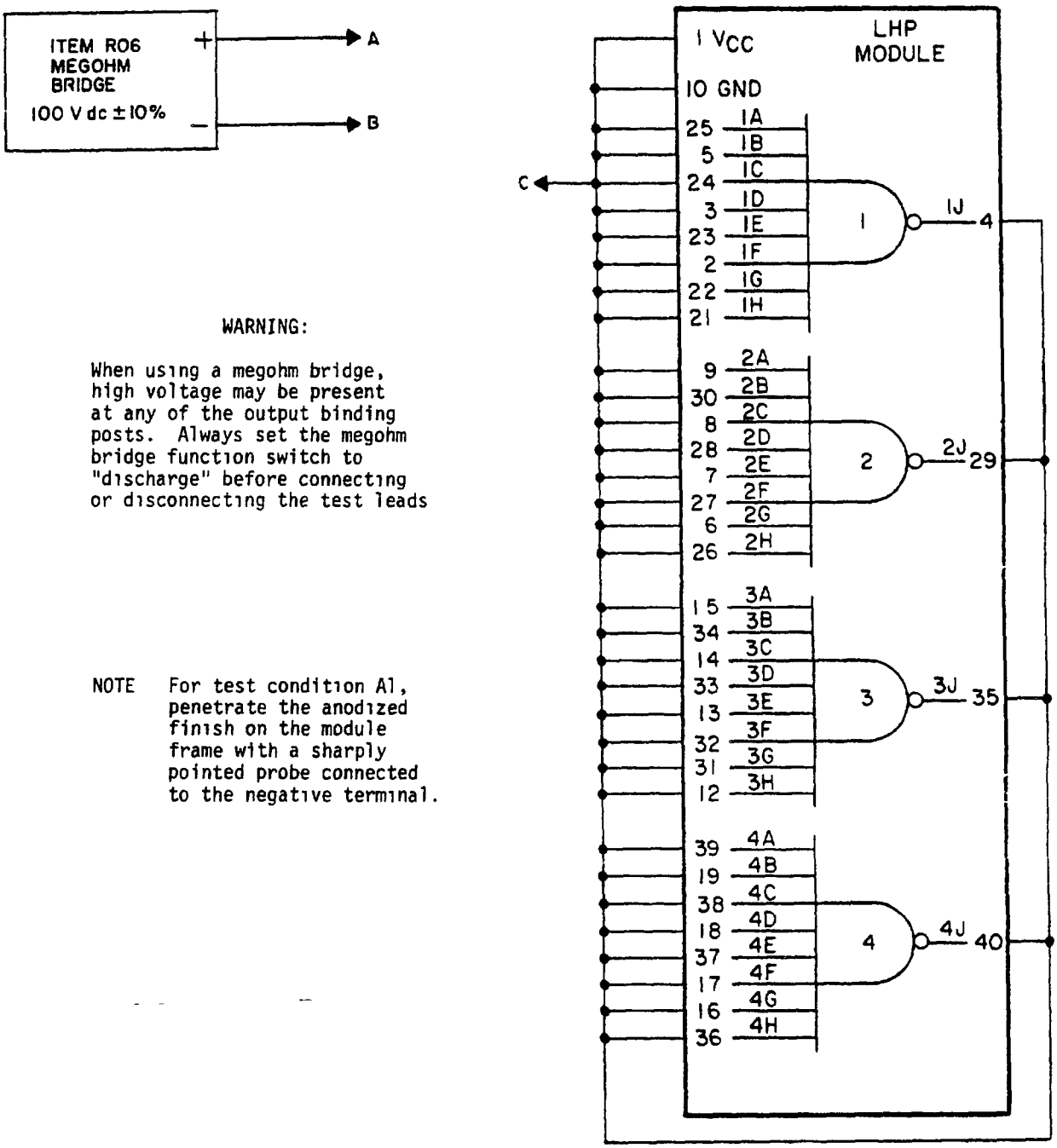
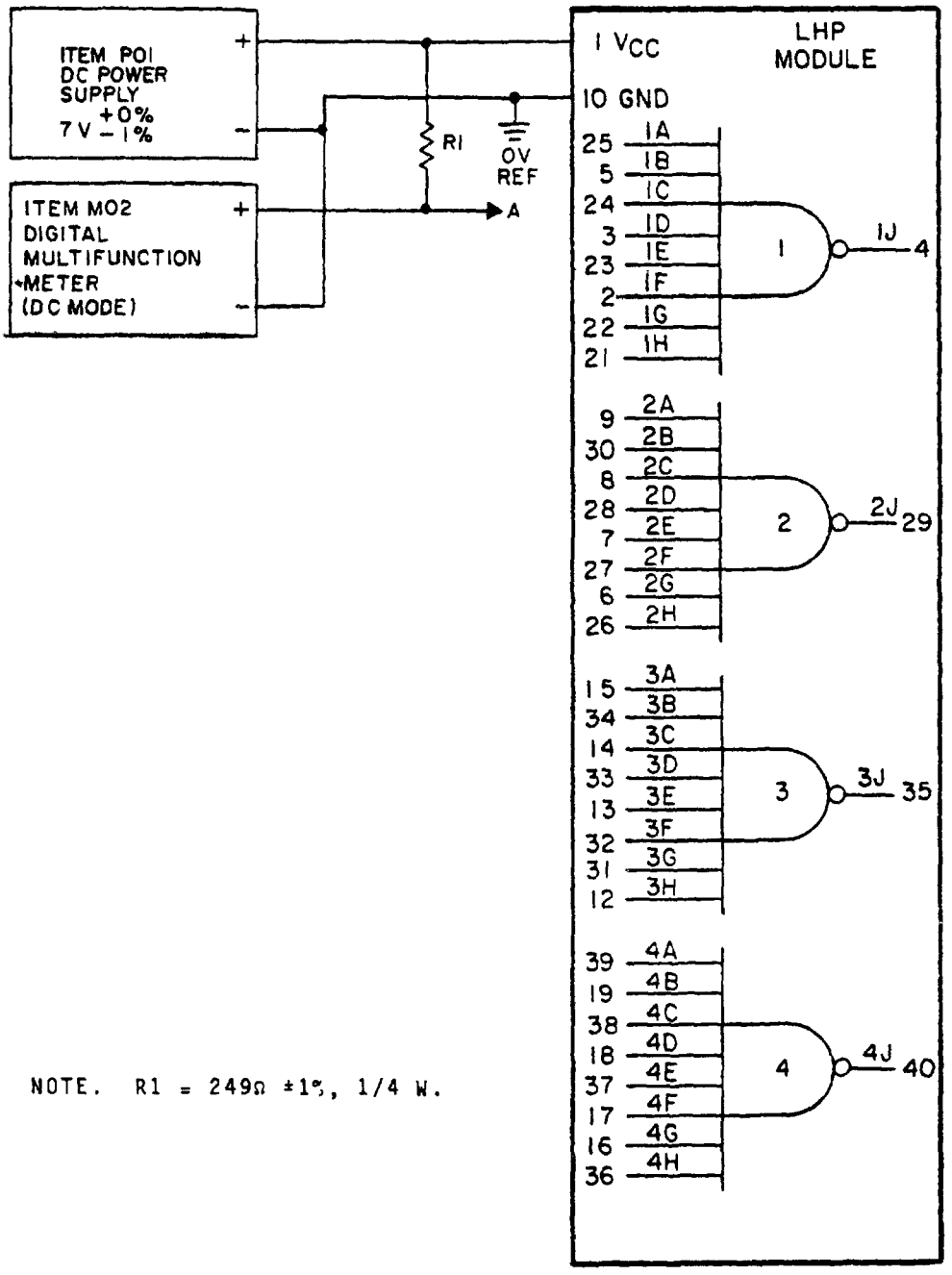


FIGURE 5. Isolation test circuit for table III

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NOTE. R1 = 249Ω ±1%, 1/4 W.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input breakdown current	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,26,27, 28,30,31,32,33,34,36,37,38,39	25	0	1	mA	7
C2	1B "	2,3,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	5	"	"	"	"
C3	1C "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,25,26, 27,28,30,31,32,33,34,36,37, 38,39	24	"	"	"	"
C4	1D "	2,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	3	"	"	"	"
C5	1E "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,24,25,26,27, 28,30,31,32,33,34,36,37,38, 39	23	"	"	"	"
C6	1F "	3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	2	"	"	"	"
C7	1G "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	22	"	"	"	"
C8	1H "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	21	"	"	"	"
C9	2A "	2,3,5,6,7,8,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	9	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C10	2B Input breakdown current	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,31,32,33,34,36,37, 38,39	30	0	1	mA	7
C11	2C "	2,3,5,6,7,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	8	"	"	"	"
C12	2D "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,30,31,32,33,34,36,37, 38,39	28	"	"	"	"
C13	2E "	2,3,5,6,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	7	"	"	"	"
C14	2F "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 28,30,31,32,33,34,36,37, 38,39	27	"	"	"	"
C15	2G "	2,3,5,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	6	"	"	"	"
C16	2H "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25, 27,28,30,31,32,33,34,36,37, 38,39	26	"	"	"	"
C17	3A "	2,3,5,6,7,8,9,12,13,14,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	15	"	"	"	"
C18	3B "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,36,37, 38,39	34	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

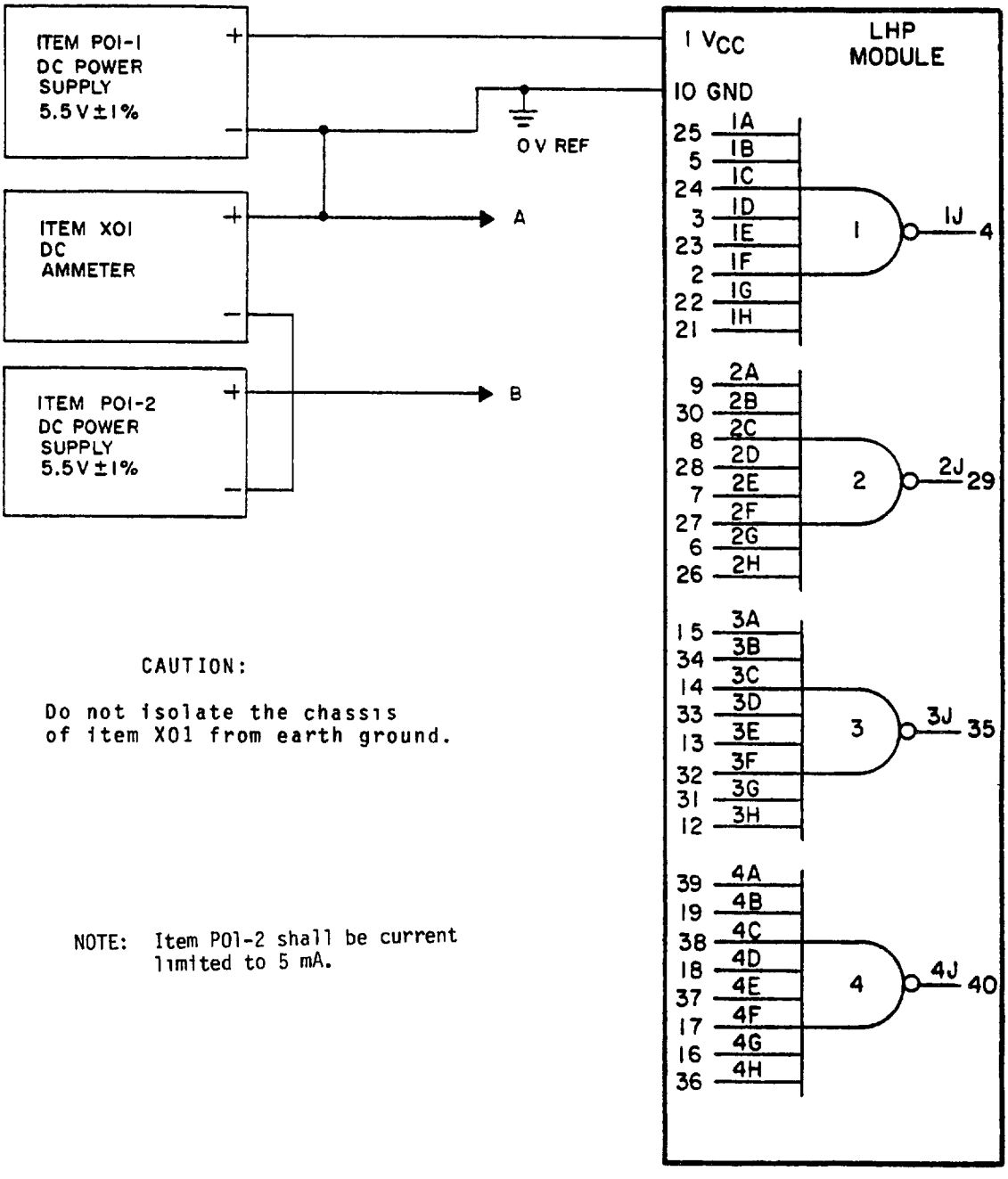
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C19	3C Input breakdown current	2,3,5,6,7,8,9,12,13,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	14	0	1	mA	7
C20	3D "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,34,36,37, 38,39	33	"	"	"	"
C21	3E "	2,3,5,6,7,8,9,12,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	13	"	"	"	"
C22	3F "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,33,34,36,37, 38,39	32	"	"	"	"
C23	3G "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,32,33,34,36,37, 38,39	31	"	"	"	"
C24	3H "	2,3,5,6,7,8,9,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	12	"	"	"	"
C25	4A "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38	39	"	"	"	"
C26	4B "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	19	"	"	"	"
C27	4C "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 39	38	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C28	4D Input breakdown current	2,3,5,6,7,8,9,12,13,14,15,16, 17,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	18	0	1	mA	7
C29	4E "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36, 38,39	37	"	"	"	"
C30	4F "	2,3,5,6,7,8,9,12,13,14,15,16, 18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	17	"	"	"	"
C31	4G "	2,3,5,6,7,8,9,12,13,14,15,17, 18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	16	"	"	"	"
C32	4H "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,37, 38,39	36	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: Item PO1-2 shall be current limited to 5 mA.

FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	1J Output short circuit current	4	-40	-100	mA	8
D2	2J "	29	"	"	"	"
D3	3J "	35	"	"	"	"
D4	4J "	40	"	"	"	"

TABLE VII. Output leakage current test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
E1	1J Output leakage current	4	0	1	mA	9
E2	2J "	29	"	"	"	"
E3	3J "	35	"	"	"	"
E4	4J "	40	"	"	"	"

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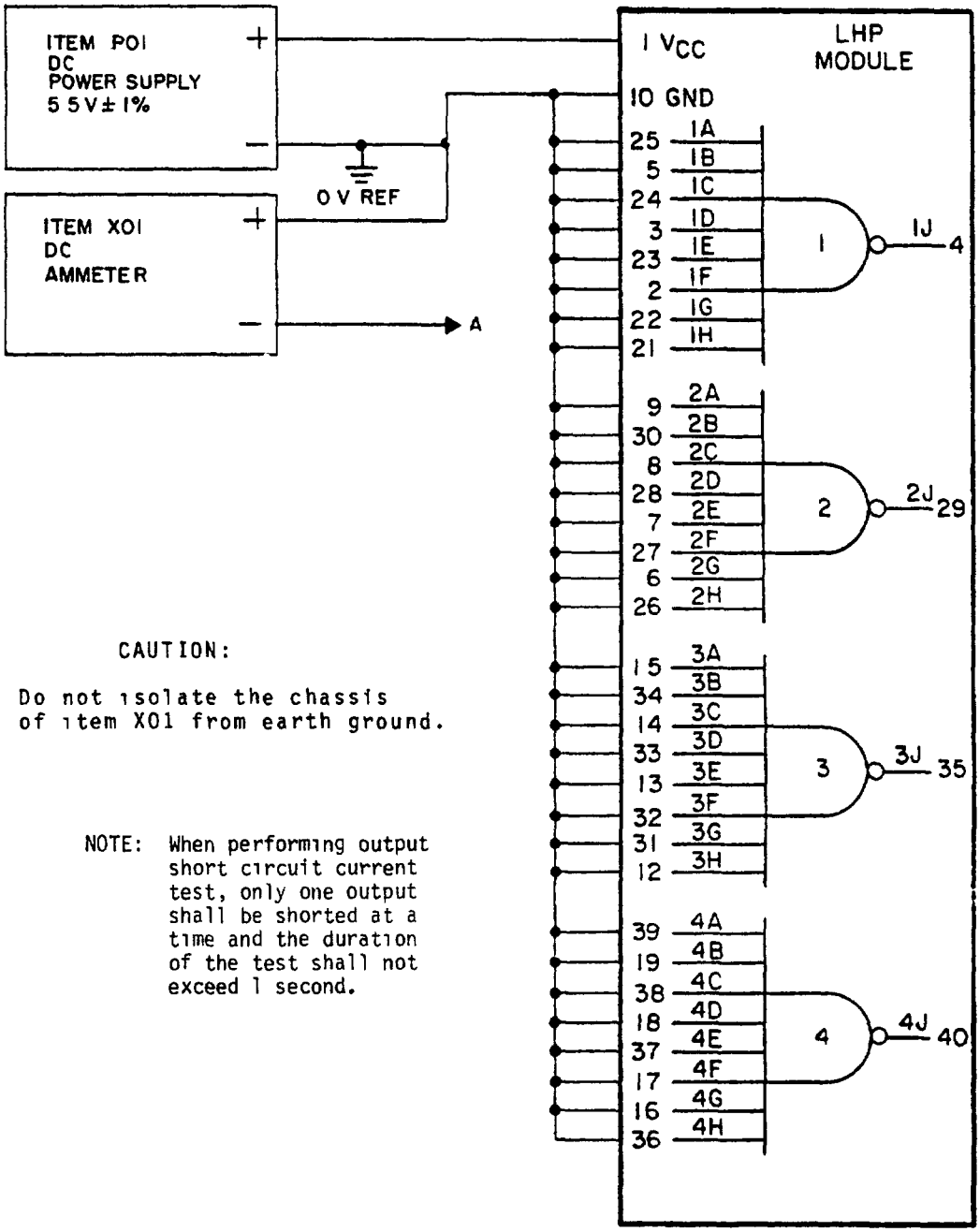


FIGURE 8. Output short circuit current test circuit for table VI

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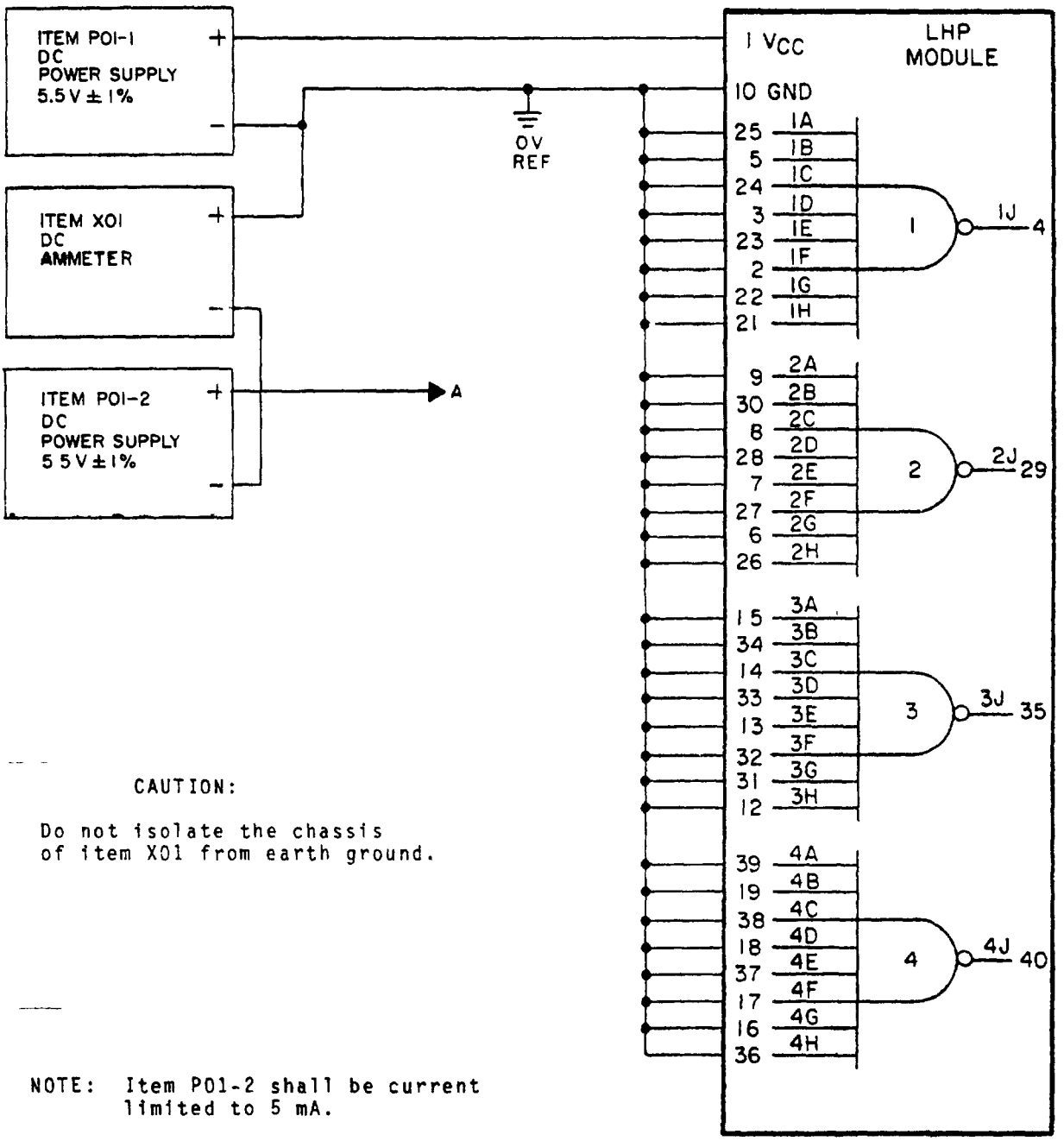


FIGURE 9. Output leakage current test circuit for table VII.

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*TABLE VIII. Input clamp diode voltage test.

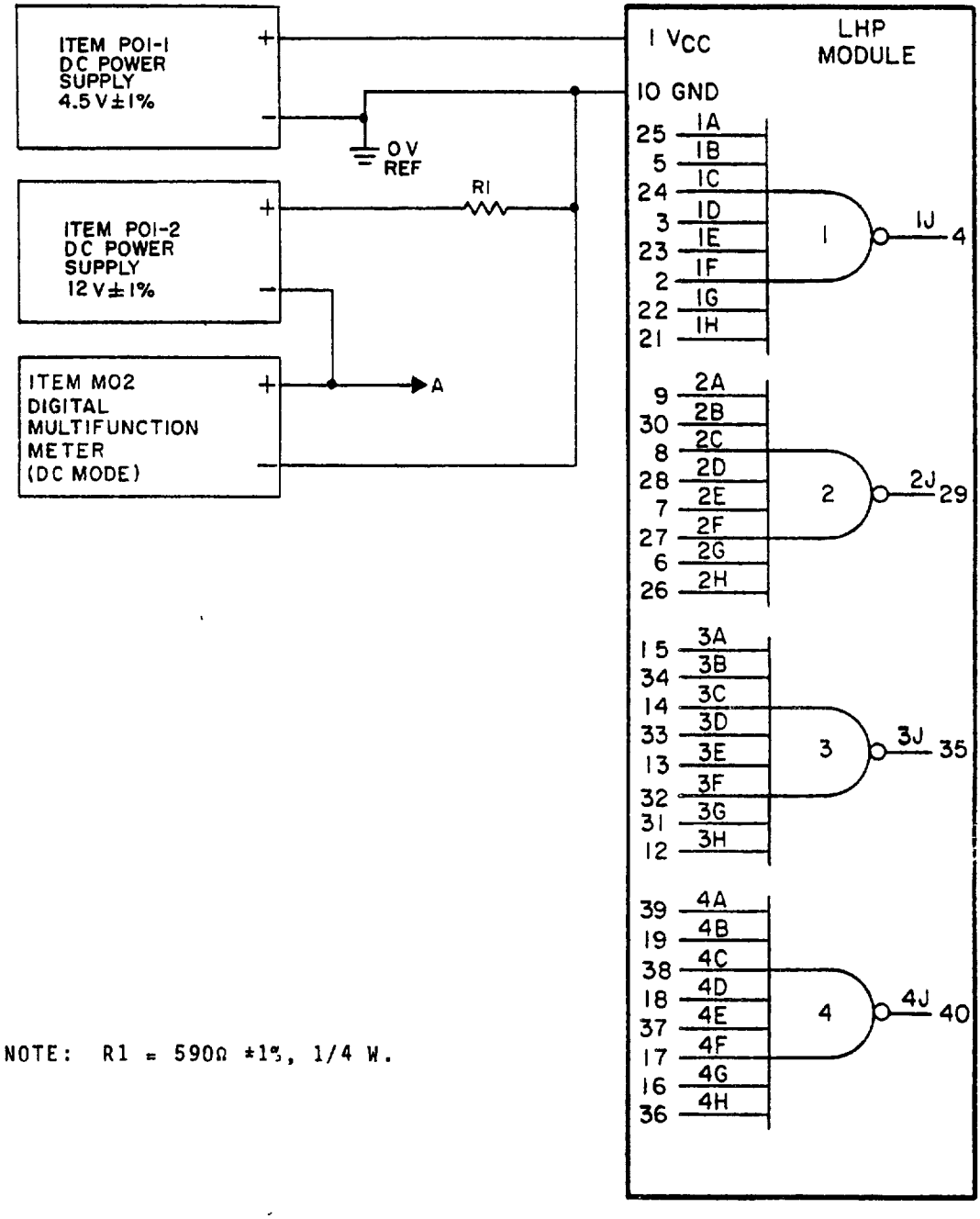
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	1A Input clamp diode voltage	25	-0.4	-1.2	V	10
F2	1B "	5	"	"	"	"
F3	1C "	24	"	"	"	"
F4	1D "	3	"	"	"	"
F5	1E "	23	"	"	"	"
F6	1F "	2	"	"	"	"
F7	1G "	22	"	"	"	"
F8	1H "	21	"	"	"	"
F9	2A "	9	"	"	"	"
F10	2B "	30	"	"	"	"
F11	2C "	8	"	"	"	"
F12	2D "	28	"	"	"	"
F13	2E "	7	"	"	"	"
F14	2F "	27	"	"	"	"
F15	2G "	6	"	"	"	"
F16	2H "	26	"	"	"	"
F17	3A "	15	"	"	"	"
F18	3B "	34	"	"	"	"
F19	3C "	14	"	"	"	"
F20	3D "	33	"	"	"	"
F21	3E "	13	"	"	"	"

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*TABLE VIII. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F22	3F Input clamp diode voltage	32	-0.4	-1.2	V	10
F23	3G "	31	"	"	"	"
F24	3H "	12	"	"	"	"
F25	4A "	39	"	"	"	"
F26	4B "	19	"	"	"	"
F27	4C "	38	"	"	"	"
F28	4D "	18	"	"	"	"
F29	4E "	37	"	"	"	"
F30	4F "	17	"	"	"	"
F31	4G "	16	"	"	"	"
F32	4H "	36	"	"	"	"

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NOTE: R1 = 590Ω ±1%, 1/4 W.

* FIGURE 10. Input clamp diode voltage test circuit for table VIII.

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*TABLE IX. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1J Output voltage, low level	25,5,24,3, 23,2,22,21	Open	4	0	500	mV	11
G2	2J "	9,30,8,28, 7,27,6,26	"	29	"	"	"	"
G3	3J "	15,34,14, 33,13,32, 31,12	"	35	"	"	"	"
G4	4J "	39,19,38, 18,37,17, 16,36	"	40	"	"	"	"

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*TABLE X. Output voltage (high-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1J Output voltage, high level	5,24,3,23, 2,22,21	25	4	2.5	4.5	V	11
H2	1J "	25,24,3,23, 2,22,21	5	"	"	"	"	"
H3	1J "	25,5,3,23, 2,22,21	24	"	"	"	"	"
H4	1J "	25,5,24,23, 2,22,21	3	"	"	"	"	"
H5	1J "	25,5,24,3, 2,22,21	23	"	"	"	"	"
H6	1J "	25,5,24,3, 23,22,21	2	"	"	"	"	"
H7	1J "	25,5,24,3, 23,2,21	22	"	"	"	"	"
H8	1J "	25,5,24,3, 23,2,22	21	"	"	"	"	"
H9	2J "	30,8,28,7, 27,6,26	9	29	"	"	"	"
H10	2J "	9,8,28,7, 27,6,26	30	"	"	"	"	"
H11	2J "	9,30,28,7, 27,6,26	8	"	"	"	"	"
H12	2J "	9,30,8,7, 27,6,26	28	"	"	"	"	"
H13	2J "	9,30,8,28, 27,6,26	7	"	"	"	"	"
H14	2J "	9,30,8,28, 7,6,26	27	"	"	"	"	"

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*TABLE X. Output voltage (high-level) test - Continued.

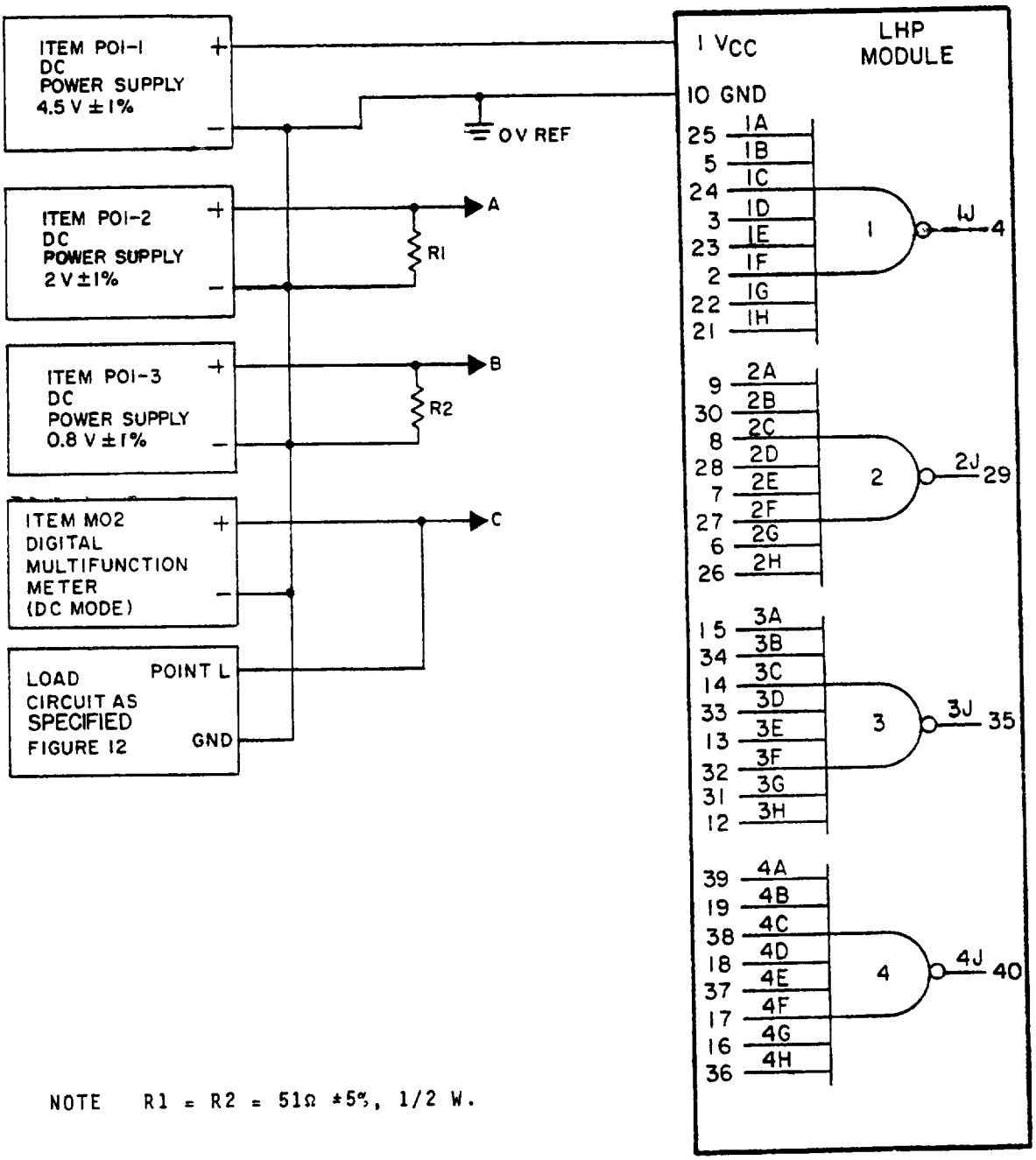
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H15	2J Output voltage, high level	9,30,8,28, 7,27,26	6	29	2.5	4.5	V	11
H16	2J "	9,30,8,28, 7,27,6	26	29	"	"	"	"
H17	3J "	34,14,33, 13,32,31,12	15	35	"	"	"	"
H18	3J "	15,14,33, 13,32,31,12	34	"	"	"	"	"
H19	3J "	15,34,33, 13,32,31,12	14	"	"	"	"	"
H20	3J "	15,34,14, 13,32,31,12	33	"	"	"	"	"
H21	3J "	15,34,14, 33,32,31,12	13	"	"	"	"	"
H22	3J "	15,34,14, 33,13,31,12	32	"	"	"	"	"
H23	3J "	15,34,14, 33,13,32,12	31	"	"	"	"	"
H24	3J "	15,34,14, 33,13,32,31	12	"	"	"	"	"
H25	4J "	19,38,18, 37,17,16,36	39	40	"	"	"	"
H26	4J "	39,38,18, 37,17,16,36	19	"	"	"	"	"
H27	4J "	39,19,18, 37,17,16,36	38	"	"	"	"	"
H28	4J "	39,19,38, 37,17,16,36	18	"	"	"	"	"

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*TABLE X. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H29	4J Output voltage, high level	39,19,38, 18,17,16,36	37	40	2.5	4.5	V	11
H30	4J "	39,19,38, 18,37,16,36	17	"	"	"	"	"
H31	4J "	39,19,38, 18,37,17,36	16	"	"	"	"	"
H32	4J "	39,19,38, 18,37,17,16	36	"	"	"	"	"

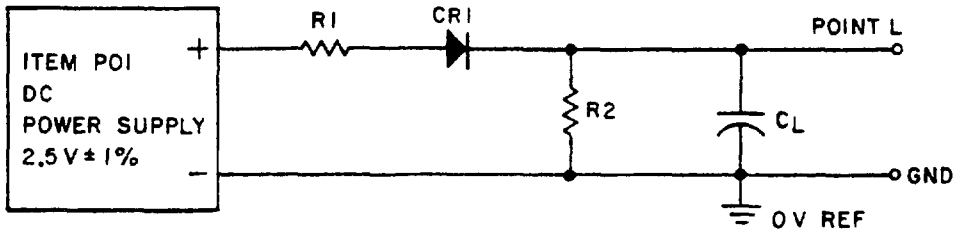
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NOTE R1 = R2 = 51Ω ±5%, 1/2 W.

* FIGURE 11. Output voltage (low-level and high-level) test circuit for tables IX and X.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 20 mA ±5% out of point L with point L held at 0.5 V.
3. R2 = 2.49 kΩ ±1%, 1/8 W.
4. Values for C_L:
   a. For test conditions M1-M4, N1-N4, P1-P32, Q1-Q32, R1-R4, and S1-S4: C_L = 50 ±5 pF (including probe and parasitic capacitance).
   b. For test conditions M5-M8, N5-N8, P33-P64, and Q33-Q64: C_L = 150 ±5 pF (including probe and parasitic capacitance).
5. Capacitor C_L is not required for test conditions G1-G4 and H1-H32.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

* FIGURE 12. LHP logic load circuit.

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TABLE XI. Input current (high-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	1A	Input current, high level	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,26,27, 28,30,31,32,33,34,36,37,38,39	25	0	50	µA	13
J2	1B	"	2,3,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	5	"	"	"	"
J3	1C	"	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,25,26, 27,28,30,31,32,33,34,36,37, 38,39	24	"	"	"	"
J4	1D	"	2,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	3	"	"	"	"
J5	1E	"	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,24,25,26,27, 28,30,31,32,33,34,36,37,38, 39	23	"	"	"	"
J6	1F	"	3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	2	"	"	"	"
J7	1G	"	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	22	"	"	"	"
J8	1H	"	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	21	"	"	"	"
J9	2A	"	2,3,5,6,7,8,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	9	"	"	"	"

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J10	2B Input current, high level	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,31,32,33,34,36,37, 38,39	30	0	50	µA	13
J11	2C "	2,3,5,6,7,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	8	"	"	"	"
J12	2D "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,30,31,32,33,34,36,37, 38,39	28	"	"	"	"
J13	2E "	2,3,5,6,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	7	"	"	"	"
J14	2F "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 28,30,31,32,33,34,36,37, 38,39	27	"	"	"	"
J15	2G "	2,3,5,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	6	"	"	"	"
J16	2H "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25, 27,28,30,31,32,33,34,36,37, 38,39	26	"	"	"	"
J17	3A "	2,3,5,6,7,8,9,12,13,14,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	15	"	"	"	"
J18	3B "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,36,37, 38,39	34	"	"	"	"

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J19	3C	Input current, high level	2,3,5,6,7,8,9,12,13,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	14	0	50	µA	13
J20	3D	"	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,34,36,37, 38,39	33	"	"	"	"
J21	3E	"	2,3,5,6,7,8,9,12,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	13	"	"	"	"
J22	3F	"	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,33,34,36,37, 38,39	32	"	"	"	"
J23	3G	"	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,32,33,34,36,37, 38,39	31	"	"	"	"
J24	3H	"	2,3,5,6,7,8,9,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	12	"	"	"	"
J25	4A	"	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38	39	"	"	"	"
J26	4B	"	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	19	"	"	"	"
J27	4C	"	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 39	38	"	"	"	"

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J28	4D Input current, high level	2,3,5,6,7,8,9,12,13,14,15,16, 17,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	18	0	50	µA	13
J29	4E "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36, 38,39	37	"	"	"	"
J30	4F "	2,3,5,6,7,8,9,12,13,14,15,16, 18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	17	"	"	"	"
J31	4G "	2,3,5,6,7,8,9,12,13,14,15,17, 18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,36,37, 38,39	16	"	"	"	"
J32	4H "	2,3,5,6,7,8,9,12,13,14,15,16, 17,18,19,21,22,23,24,25,26, 27,28,30,31,32,33,34,37, 38,39	36	"	"	"	"

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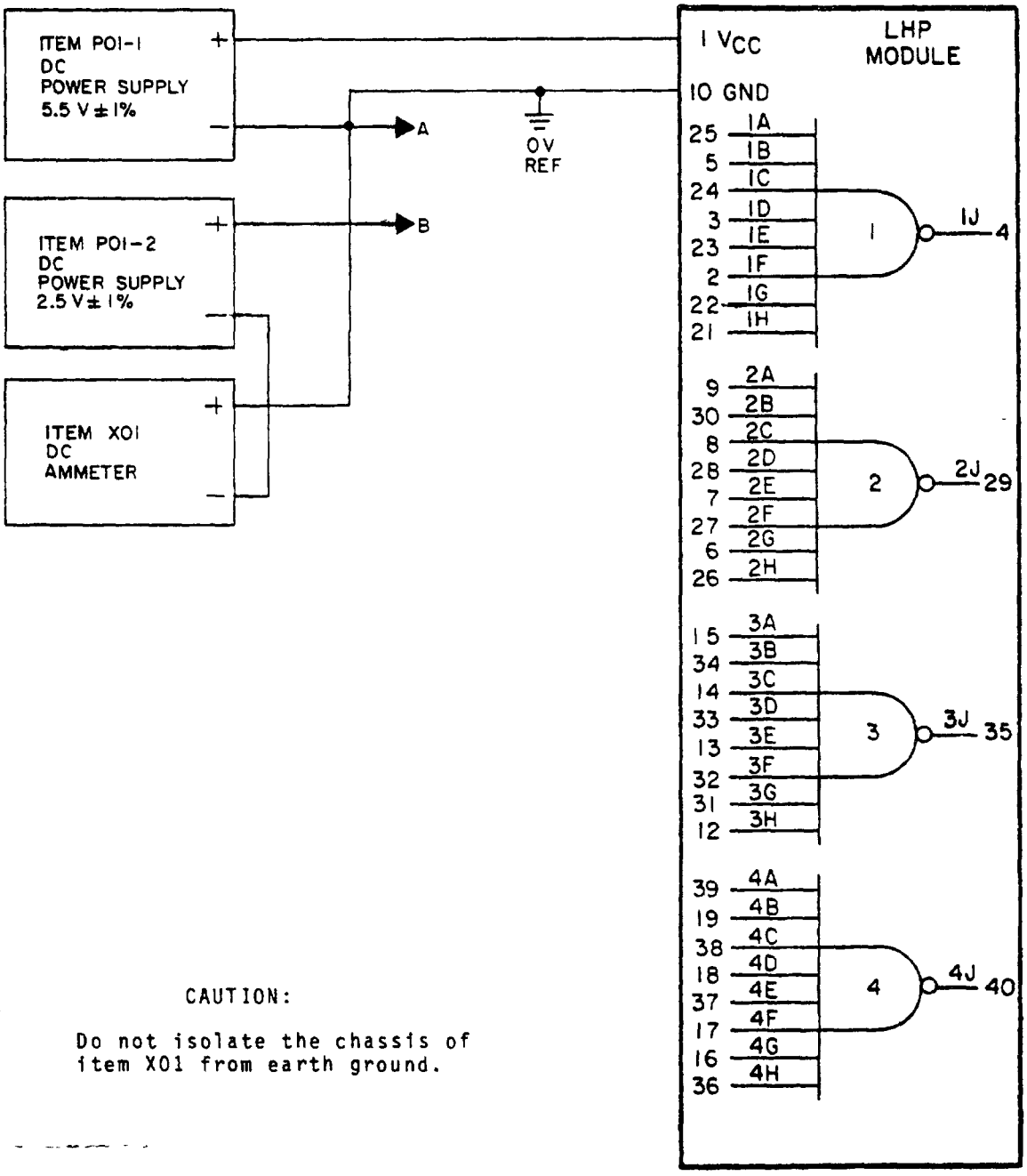


FIGURE 13. Input current (high-level) test circuit for table XI.

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TABLE XII. Input current (low-level) test.

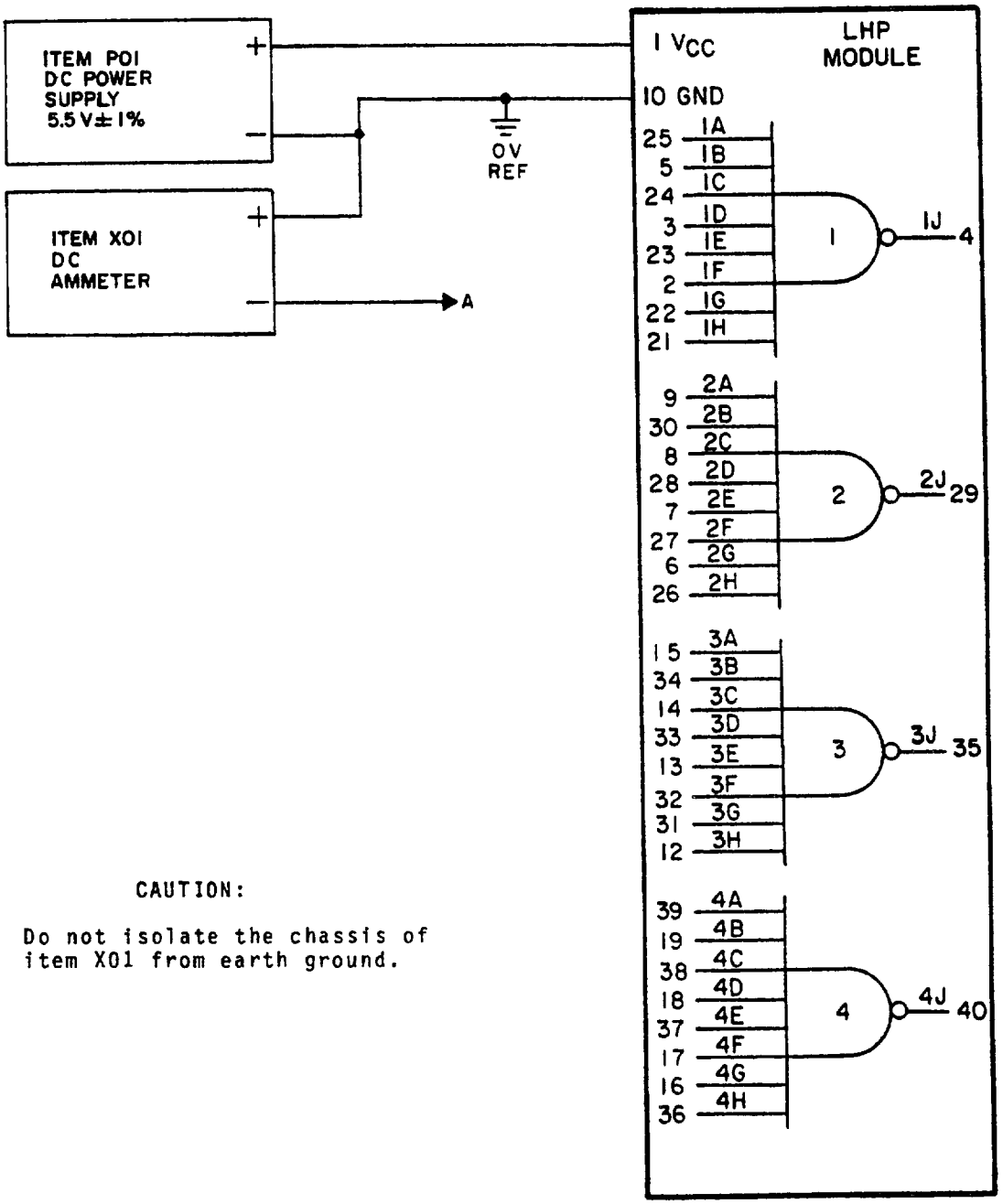
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
K1	1A Input current, low level	25	-0.4	-2.2	mA	14
K2	1B "	5	"	"	"	"
K3	1C "	24	"	"	"	"
K4	1D "	3	"	"	"	"
K5	1E "	23	"	"	"	"
K6	1F "	2	"	"	"	"
K7	1G "	22	"	"	"	"
K8	1H "	21	"	"	"	"
K9	2A "	9	"	"	"	"
K10	2B "	30	"	"	"	"
K11	2C "	8	"	"	"	"
K12	2D "	28	"	"	"	"
K13	2E "	7	"	"	"	"
K14	2F "	27	"	"	"	"
K15	2G "	6	"	"	"	"
K16	2H "	26	"	"	"	"
K17	3A "	15	"	"	"	"
K18	3B "	34	"	"	"	"
K19	3C "	14	"	"	"	"
K20	3D "	33	"	"	"	"
K21	3E "	13	"	"	"	"
K22	3F "	32	"	"	"	"

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TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K23	3G Input current, low level	31	-0.4	-2.2	mA	14
K24	3H "	12	"	"	"	"
K25	4A "	39	"	"	"	"
K26	4B "	19	"	"	"	"
K27	4C "	38	"	"	"	"
K28	4D "	18	"	"	"	"
K29	4E "	37	"	"	"	"
K30	4F "	17	"	"	"	"
K31	4G "	16	"	"	"	"
K32	4H "	36	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 14. Input current (low-level) test circuit for table XII.

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TABLE XIII. Power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Power supply current, low level	C	Open	5	40	mA	15
L2	Power supply current, high level	Open	C	2	20	mA	15

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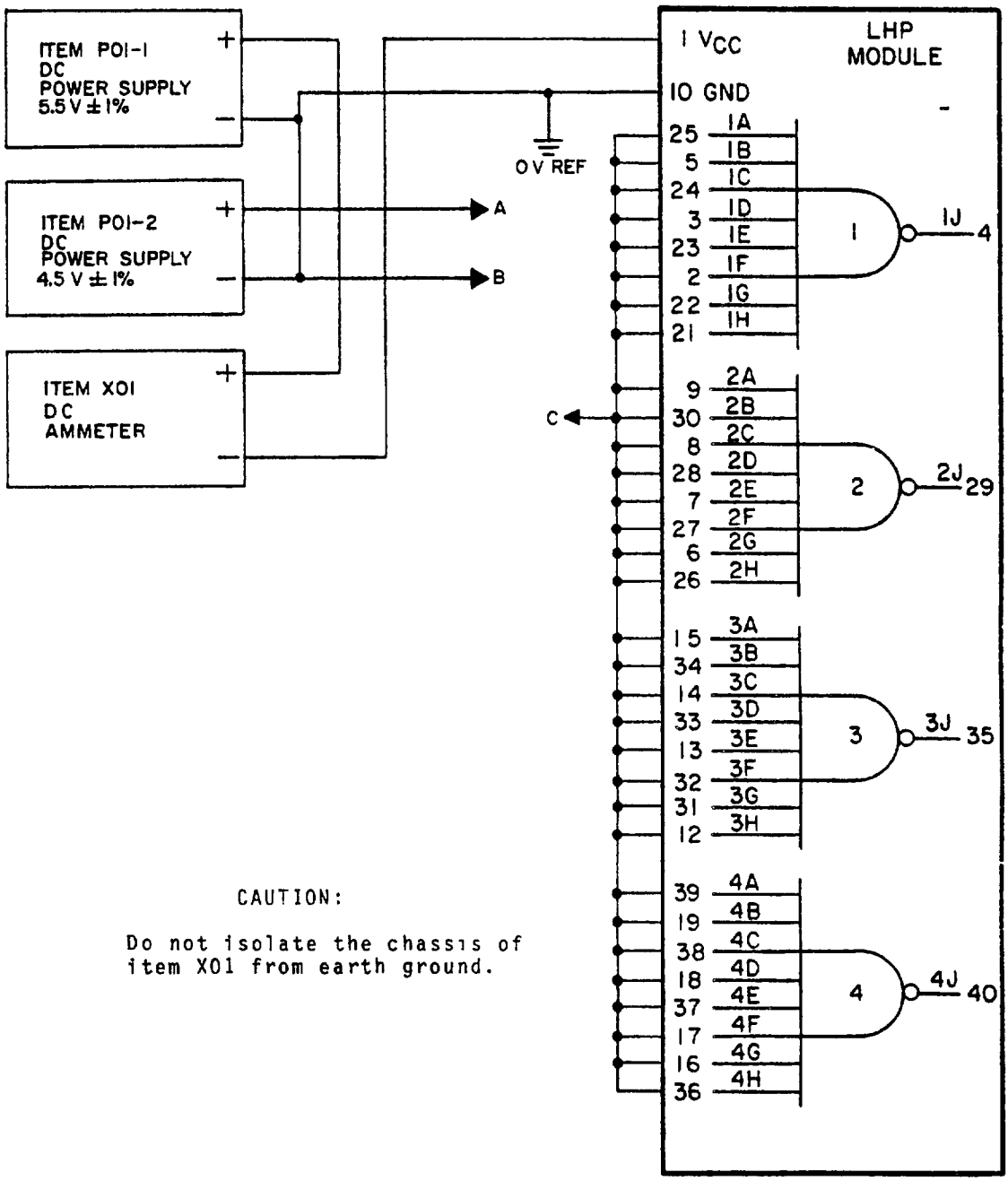


FIGURE 15. Power supply current (low-level and high-level) test circuit for table XIII.

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	1J Transition time (t_{TLH}), C_L = 50 pF	5,24,3,23, 2,22,21	25	4	2	15	ns	16,17
M2	2J "	30,8,28,7, 27,6,26	9	29	"	"	"	"
M3	3J "	34,14,33, 13,32,31,12	15	35	"	"	"	"
M4	4J "	19,38,18, 37,17,16,36	39	40	"	"	"	"
M5	1J Transition time (t_{TLH}), C_L = 150 pF	5,24,3,23, 2,22,21	25	4	6	27	"	"
M6	2J "	30,8,28,7, 27,6,26	9	29	"	"	"	"
M7	3J "	34,14,33, 13,32,31,12	15	35	"	"	"	"
M8	4J "	19,38,18, 37,17,16,36	39	40	"	"	"	"
N1	1J Transition time (t_{THL}), C_L = 50 pF	5,24,3,23, 2,22,21	25	4	2	6	"	"
N2	2J "	30,8,28,7, 27,6,26	9	29	"	"	"	"
N3	3J "	34,14,33, 13,32,31,12	15	35	"	"	"	"
N4	4J "	19,38,18, 37,17,16,36	39	40	"	"	"	"
N5	1J Transition time (t_{THL}), C_L = 150 pF	5,24,3,23, 2,22,21	25	4	3	11	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
N6	2J	Transition time (t_{THL}), C_L = 150 pF	30,8,28,7, 27,6,26	9	29	3	11	ns	16,17
N7	3J	"	34,14,33, 13,32,31,12	15	35	"	"	"	"
N8	4J	"	19,38,18, 37,17,16,36	39	40	"	"	"	"
P1	1A to 1J	Propagation delay time (t_{PLH}), C_L = 50 pF	5,24,3,23, 2,22,21	25	4	2	8	"	"
P2	1B to 1J	"	25,24,3,23, 2,22,21	5	"	"	"	"	"
P3	1C to 1J	"	25,5,3,23, 2,22,21	24	"	"	"	"	"
P4	1D to 1J	"	25,5,24,23, 2,22,21	3	"	"	"	"	"
P5	1E to 1J	"	25,5,24,3, 2,22,21	23	"	"	"	"	"
P6	1F to 1J	"	25,5,24,3, 23,22,21	2	"	"	"	"	"
P7	1G to 1J	"	25,5,24,3, 23,2,21	22	"	"	"	"	"
P8	1H to 1J	"	25,5,24,3, 23,2,22	21	"	"	"	"	"
P9	2A to 2J	"	30,8,28,7, 27,6,26	9	29	"	"	"	"
P10	2B to 2J	"	9,8,28,7, 27,6,26	30	"	"	"	"	"
P11	2C to 2J	"	9,30,28,7, 27,6,26	8	"	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P12	2D to 2J	Propagation delay time (t_{PLH}), C_L = 50 pF	9,30,8,7, 27,6,26	28	29	2	8	ns	16,17
P13	2E to 2J	"	9,30,8,28, 27,6,26	7	"	"	"	"	"
P14	2F to 2J	"	9,30,8,28, 7,6,26	27	"	"	"	"	"
P15	2G to 2J	"	9,30,8,28, 7,27,26	6	"	"	"	"	"
P16	2H to 2J	"	9,30,8,28, 7,27,6	26	"	"	"	"	"
P17	3A to 3J	"	34,14,33, 13,32,31,12	15	35	"	"	"	"
P18	3B to 3J	"	15,14,33, 13,32,31,12	34	"	"	"	"	"
P19	3C to 3J	"	15,34,33, 13,32,31,12	14	"	"	"	"	"
P20	3D to 3J	"	15,34,14, 13,32,31,12	33	"	"	"	"	"
P21	3E to 3J	"	15,34,14, 33,32,31,12	13	"	"	"	"	"
P22	3F to 3J	"	15,34,14, 33,13,31,12	32	"	"	"	"	"
P23	3G to 3J	"	15,34,14, 33,13,32,12	31	"	"	"	"	"
P24	3H to 3J	"	15,34,14, 33,13,32,31	12	"	"	"	"	"
P25	4A to 4J	"	19,38,18, 37,17,16,36	39	40	"	"	"	"
P26	4B to 4J	"	39,38,18, 37,17,16,36	19	40	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P27	4C to 4J	Propagation delay time (t_{PLH}), C_L = 50 pF	39,19,18, 37,17,16,36	38	40	2	8	ns	16,17
P28	4D to 4J	"	39,19,38, 37,17,16,36	18	"	"	"	"	"
P29	4E to 4J	"	39,19,38, 18,17,16,36	37	"	"	"	"	"
P30	4F to 4J	"	39,19,38, 18,37,16,36	17	"	"	"	"	"
P31	4G to 4J	"	39,19,38, 18,37,17,36	16	"	"	"	"	"
P32	4H to 4J	"	39,19,38, 18,37,17,16	36	"	"	"	"	"
P33	1A to 1J	Propagation delay time (t_{PLH}), C_L = 150 pF	5,24,3,23, 2,22,21	25	4	3	12	"	"
P34	1B to 1J	"	25,24,3,23, 2,22,21	5	"	"	"	"	"
P35	1C to 1J	"	25,5,3,23, 2,22,21	24	"	"	"	"	"
P36	1D to 1J	"	25,5,24,23, 2,22,21	3	"	"	"	"	"
P37	1E to 1J	"	25,5,24,3, 2,22,21	23	"	"	"	"	"
P38	1F to 1J	"	25,5,24,3, 23,22,21	2	"	"	"	"	"
P39	1G to 1J	"	25,5,24,3, 23,2,21	22	"	"	"	"	"
P40	1H to 1J	"	25,5,24,3, 23,2,22	21	"	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P41	2A to 2J	Propagation delay time (t_{PLH}), C_L = 150 pF	30,8,28,7, 27,6,26	9	29	3	12	ns	16,17
P42	2B to 2J	"	9,8,28,7, 27,6,26	30	"	"	"	"	"
P43	2C to 2J	"	9,30,28,7, 27,6,26	8	"	"	"	"	"
P44	2D to 2J	"	9,30,8,7, 27,6,26	28	"	"	"	"	"
P45	2E to 2J	"	9,30,8,28, 27,6,26	7	"	"	"	"	"
P46	2F to 2J	"	9,30,8,28, 7,6,26	27	"	"	"	"	"
P47	2G to 2J	"	9,30,8,28, 7,27,26	6	"	"	"	"	"
P48	2H to 2J	"	9,30,8,28, 7,27,6	26	"	"	"	"	"
P49	3A to 3J	"	34,14,33, 13,32,31,12	15	35	"	"	"	"
P50	3B to 3J	"	15,14,33, 13,32,31,12	34	"	"	"	"	"
P51	3C to 3J	"	15,34,33, 13,32,31,12	14	"	"	"	"	"
P52	3D to 3J	"	15,34,14, 13,32,31,12	33	"	"	"	"	"
P53	3E to 3J	"	15,34,14, 33,32,31,12	13	"	"	"	"	"
P54	3F to 3J	"	15,34,14, 33,13,31,12	32	"	"	"	"	"
P55	3G to 3J	"	15,34,14, 33,13,32,12	31	"	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P56	3H to 3J	Propagation delay time (t_{PLH}), C_L = 150 pF	15,34,14, 33,13,32,31	12	35	3	12	ns	16,17
P57	4A to 4J	"	19,38,18, 37,17,16,36	39	40	"	"	"	"
P58	4B to 4J	"	39,38,18, 37,17,16,36	19	"	"	"	"	"
P59	4C to 4J	"	39,19,18, 37,17,16,36	38	"	"	"	"	"
P60	4D to 4J	"	39,19,38, 37,17,16,36	18	"	"	"	"	"
P61	4E to 4J	"	39,19,38, 18,17,16,36	37	"	"	"	"	"
P62	4F to 4J	"	39,19,38, 18,37,16,36	17	"	"	"	"	"
P63	4G to 4J	"	39,19,38, 18,37,17,36	16	"	"	"	"	"
P64	4H to 4J	"	39,19,38, 18,37,17,16	36	"	"	"	"	"
Q1	1A to 1J	Propagation delay time (t_{PHL}), C_L = 50 pF	5,24,3,23, 2,22,21	25	4	2	9	"	"
Q2	1B to 1J	"	25,24,3,23, 2,22,21	5	"	"	"	"	"
Q3	1C to 1J	"	25,5,3,23, 2,22,21	24	"	"	"	"	"
Q4	1D to 1J	"	25,5,24,23, 2,22,21	3	"	"	"	"	"
Q5	1E to 1J	"	25,5,24,3, 2,22,21	23	"	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q6	1F to 1J	Propagation delay time (t_{PHL}), C_L = 50 pF	25,5,24,3, 23,22,21	2	4	2	9	ns	16,17
Q7	1G to 1J	"	25,5,24,3, 23,2,21	22	"	"	"	"	
Q8	1H to 1J	"	25,5,24,3, 23,2,22	21	"	"	"	"	"
Q9	2A to 2J	"	30,8,28,7, 27,6,26	9	29	"	"	"	"
Q10	2B to 2J	"	9,8,28,7, 27,6,26	30	"	"	"	"	"
Q11	2C to 2J	"	9,30,28,7, 27,6,26	8	"	"	"	"	"
Q12	2D to 2J	"	9,30,8,7, 27,6,26	28	"	"	"	"	"
Q13	2E to 2J	"	9,30,8,28, 27,6,26	7	"	"	"	"	"
Q14	2F to 2J	"	9,30,8,28, 7,6,26	27	"	"	"	"	"
Q15	2G to 2J	"	9,30,8,28, 7,27,26	6	"	"	"	"	"
Q16	2H to 2J	"	9,30,8,28, 7,27,6	26	"	"	"	"	"
Q17	3A to 3J	"	34,14,33, 13,32,31,12	15	35	"	"	"	"
Q18	3B to 3J	"	15,14,33, 13,32,31,12	34	"	"	"	"	"
Q19	3C to 3J	"	15,34,33, 13,32,31,12	14	"	"	"	"	"
Q20	3D to 3J	"	15,34,14, 13,32,31,12	33	"	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q21	3E to 3J	Propagation delay time (t_{PHL}), C_L = 50 pF	15,34,14, 33,32,31,12	13	35	2	9	ns	16,17
Q22	3F to 3J	"	15,34,14, 33,13,31,12	32	"	"	"	"	"
Q23	3G to 3J	"	15,34,14, 33,13,32,12	31	"	"	"	"	"
Q24	3H to 3J	"	15,34,14, 33,13,32,31	12	"	"	"	"	"
Q25	4A to 4J	"	19,38,18, 37,17,16,36	39	40	"	"	"	"
Q26	4B to 4J	"	39,38,18, 37,17,16,36	19	"	"	"	"	"
Q27	4C to 4J	"	39,19,18, 37,17,16,36	38	"	"	"	"	"
Q28	4D to 4J	"	39,19,38, 37,17,16,36	18	"	"	"	"	"
Q29	4E to 4J	"	39,19,38, 18,17,16,36	37	"	"	"	"	"
Q30	4F to 4J	"	39,19,38, 18,37,16,36	17	"	"	"	"	"
Q31	4G to 4J	"	39,19,38, 18,37,17,36	16	"	"	"	"	"
Q32	4H to 4J	"	39,19,38, 18,37,17,16	36	"	"	"	"	"
Q33	1A to 1J	Propagation delay time (t_{PHL}), C_L = 150 pF	5,24,3,23, 2,22,21	25	4	3	13	"	"
Q34	1B to 1J	"	25,24,3,23, 2,22,21	5	"	"	"	"	"
Q35	1C to 1J	"	25,5,3,23, 2,22,21	24	"	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q36	1D to 1J	Propagation delay time (t_{PHL}), C_L = 150 pF	25,5,24,23, 2,22,21	3	4	3	13	ns	16,17
Q37	1E to 1J	"	25,5,24,3, 2,22,21	23	"	"	"	"	"
Q38	1F to 1J	"	25,5,24,3, 23,22,21	2	"	"	"	"	"
Q39	1G to 1J	"	25,5,24,3, 23,2,21	22	"	"	"	"	"
Q40	1H to 1J	"	25,5,24,3, 23,2,22	21	"	"	"	"	"
Q41	2A to 2J	"	30,8,28,7, 27,6,26	9	29	"	"	"	"
Q42	2B to 2J	"	9,8,28,7, 27,6,26	30	"	"	"	"	"
Q43	2C to 2J	"	9,30,28,7, 27,6,26	8	"	"	"	"	"
Q44	2D to 2J	"	9,30,8,7, 27,6,26	28	"	"	"	"	"
Q45	2E to 2J	"	9,30,8,28, 27,6,26	7	"	"	"	"	"
Q46	2F to 2J	"	9,30,8,28, 7,6,26	27	"	"	"	"	"
Q47	2G to 2J	"	9,30,8,28, 7,27,26	6	"	"	"	"	"
Q48	2H to 2J	"	9,30,8,28, 7,27,6	26	"	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

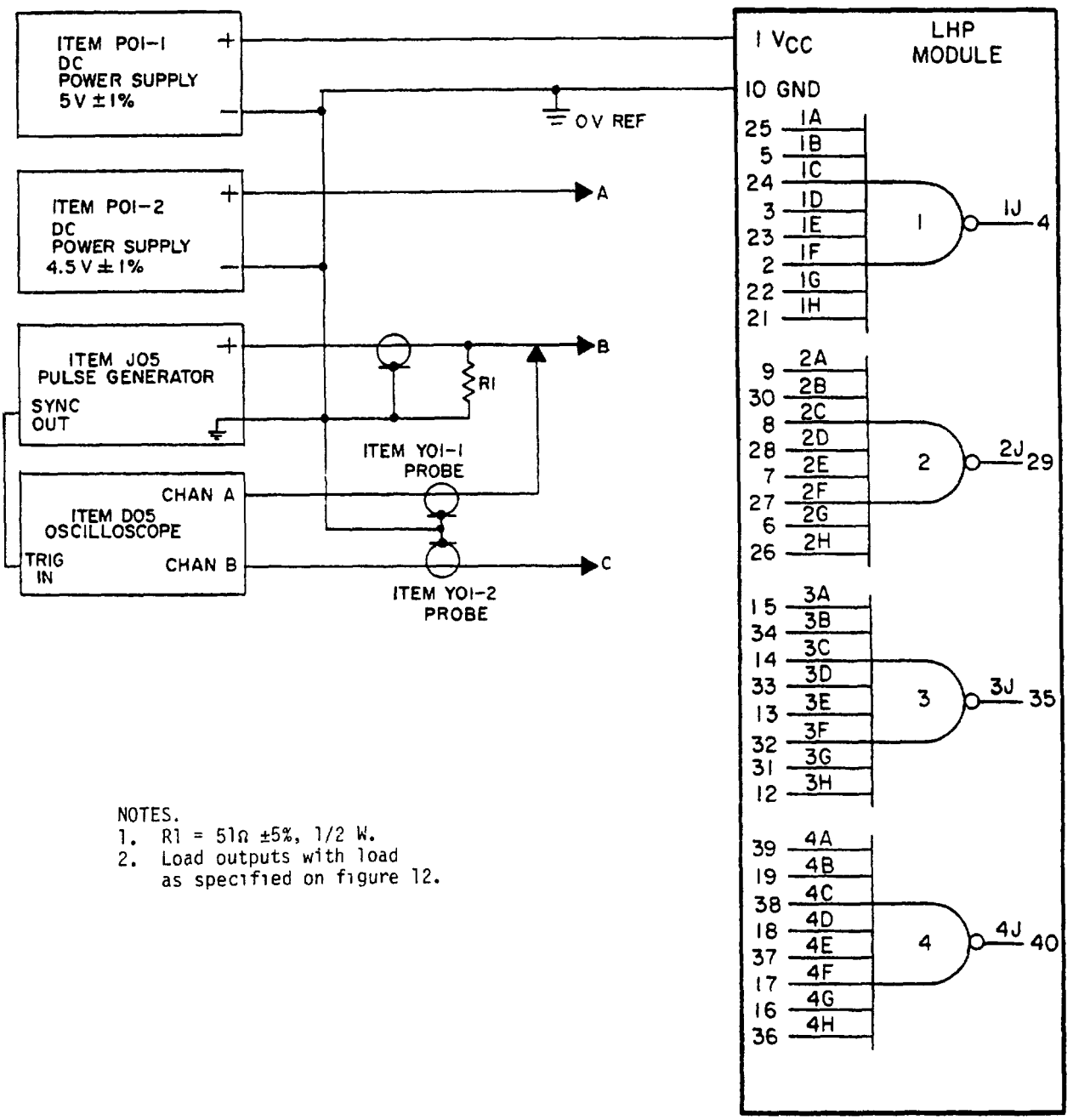
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q49	3A to 3J	Propagation delay time (t_{PHL}), C_L = 150 pF	34,14,33, 13,32,31,12	15	35	3	13	ns	16,17
Q50	3B to 3J	"	15,14,33, 13,32,31,12	34	"	"	"	"	"
Q51	3C to 3J	"	15,34,33, 13,32,31,12	14	"	"	"	"	"
Q52	3D to 3J	"	15,34,14, 13,32,31,12	33	"	"	"	"	"
Q53	3E to 3J	"	15,34,14, 33,32,31,12	13	"	"	"	"	"
Q54	3F to 3J	"	15,34,14, 33,13,31,12	32	"	"	"	"	"
Q55	3G to 3J	"	15,34,14, 33,13,32,12	31	"	"	"	"	"
Q56	3H to 3J	"	15,34,14, 33,13,32,31	12	"	"	"	"	"
Q57	4A to 4J	"	19,38,18, 37,17,16,36	39	40	"	"	"	"
Q58	4B to 4J	"	39,38,18, 37,17,16,36	19	"	"	"	"	"
Q59	4C to 4J	"	39,19,18, 37,17,16,36	38	"	"	"	"	"
Q60	4D to 4J	"	39,19,38, 37,17,16,36	18	"	"	"	"	"
Q61	4E to 4J	"	39,19,38, 18,17,16,36	37	"	"	"	"	"

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*TABLE XIV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q62	4F to 4J	Propagation delay time (t_{PHL}), C_L = 150 pF	39,19,38, 18,37,16,36	17	40	3	13	ns	16,17
Q63	4G to 4J	"	39,19,38, 18,37,17,36	16	"	"	"	"	"
Q64	4H to 4J	"	39,19,38, 18,37,17,16	36	"	"	"	"	"

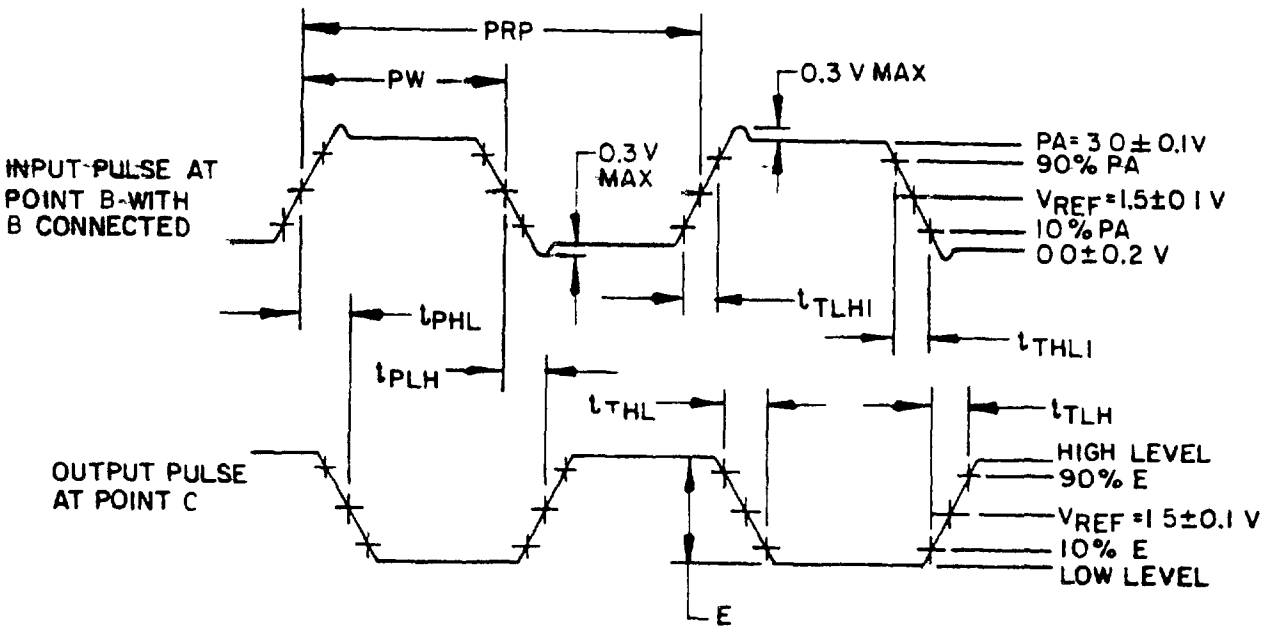
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NOTES.
1. R1 = 51Ω ±5%, 1/2 W.
2. Load outputs with load as specified on figure 12.

* FIGURE 16. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XIV.

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NOTES.
1. Input pulse parameters:
   $t_{TLHI} = t_{THLI} = 4 \pm 1$ ns.
   PW = 100 ±10 ns.
   PRP = 500 ±50 ns.
2. Before making the measurements, verify that the output pulse high level is ≥ 2.5 V and the low level is ≤ 0.5 V.
3. The lead length of electrical conductors between resistor R1 and input pins, between items YO1-1 and YO1-2 and input and output pins, and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

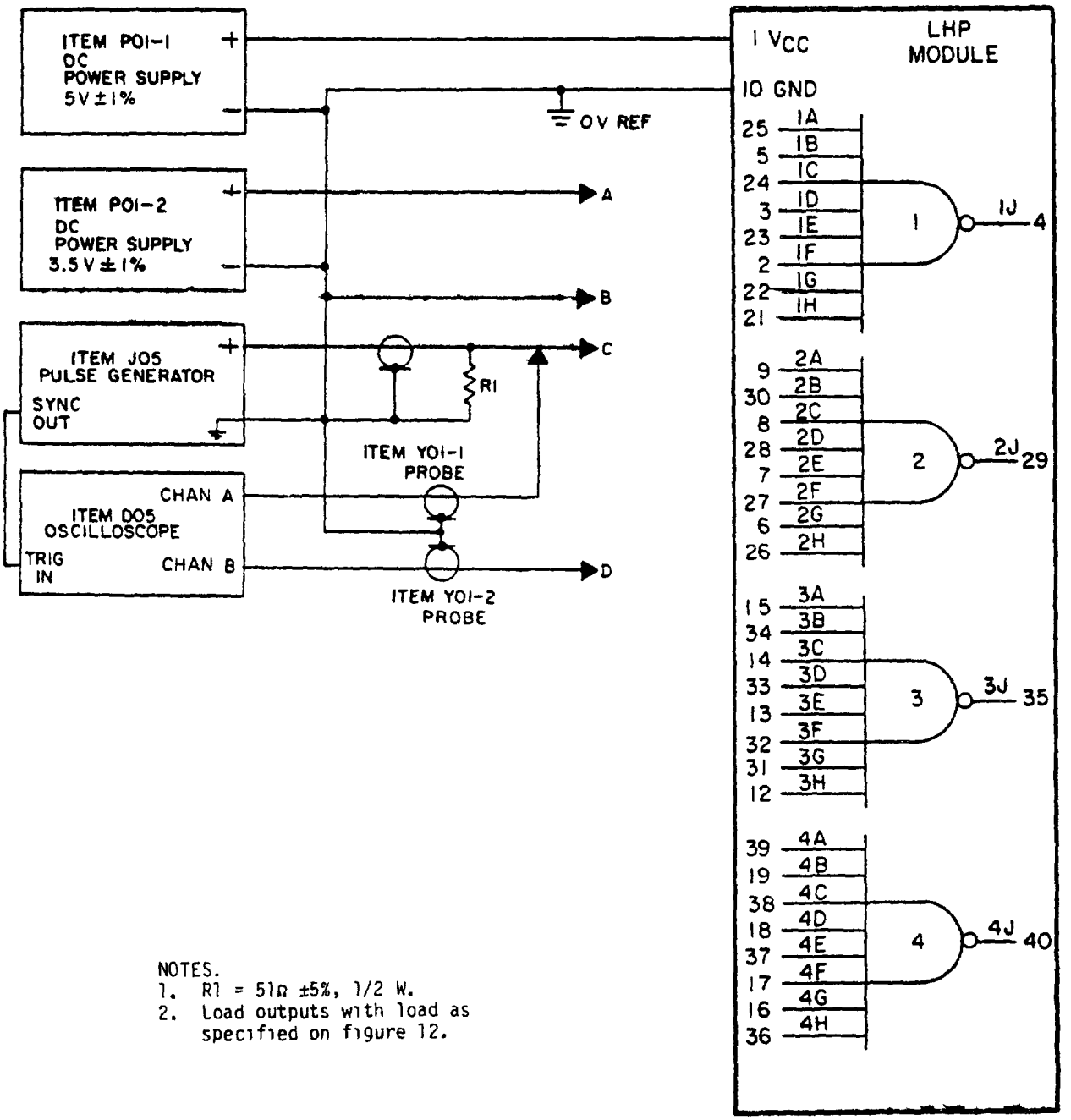
* FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XIV.

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TABLE XV. Output crosstalk (low-level and high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1J Output crosstalk, low level	25,5,24,3,23,2,22,21,30, 8,28,7,27,6,26,34,14,33, 13,32,31,12,19,38,18,37, 17,16,36	Open	9,15,39	4	0	500	mV	18,19
R2	2J "	9,30,8,28,7,27,6,26,5, 24,3,23,2,22,21,34,14, 33,13,32,31,12,19,38,18, 37,17,16,36	"	25,15,39	29	"	"	"	"
R3	3J "	15,34,14,33,13,32,31,12, 5,24,3,23,2,22,21,30,8, 28,7,27,6,26,19,38,18, 37,17,16,36	"	25,9,39	35	"	"	"	"
R4	4J "	39,19,38,18,37,17,16,36, 5,24,3,23,2,22,21,30,8, 28,7,27,6,26,34,14,33, 13,32,31,12	"	25,9,15	40	"	"	"	"
S1	1J Output crosstalk, high level	30,8,28,7,27,6,26,34,14, 33,13,32,31,12,19,38,18, 37,17,16,36	25,4,24,3, 23,2,22,21	9,15,39	4	"	400	"	"
S2	2J "	5,24,3,23,2,22,21,34,14, 33,13,32,31,12,19,38,18, 37,17,16,36	9,30,8,28, 7,27,6,26	25,15,39	29	"	"	"	"
S3	3J "	5,24,3,23,2,22,21,30,8, 28,7,27,6,26,19,38,18, 37,17,16,36	15,34,14, 33,13,32, 31,12	25,9,39	35	"	"	"	"
S4	4J "	5,24,3,23,2,22,21,30,8, 28,7,27,6,26,34,14,33, 13,32,31,12	39,19,38, 18,37,17, 16,36	25,9,15	40	"	"	"	"

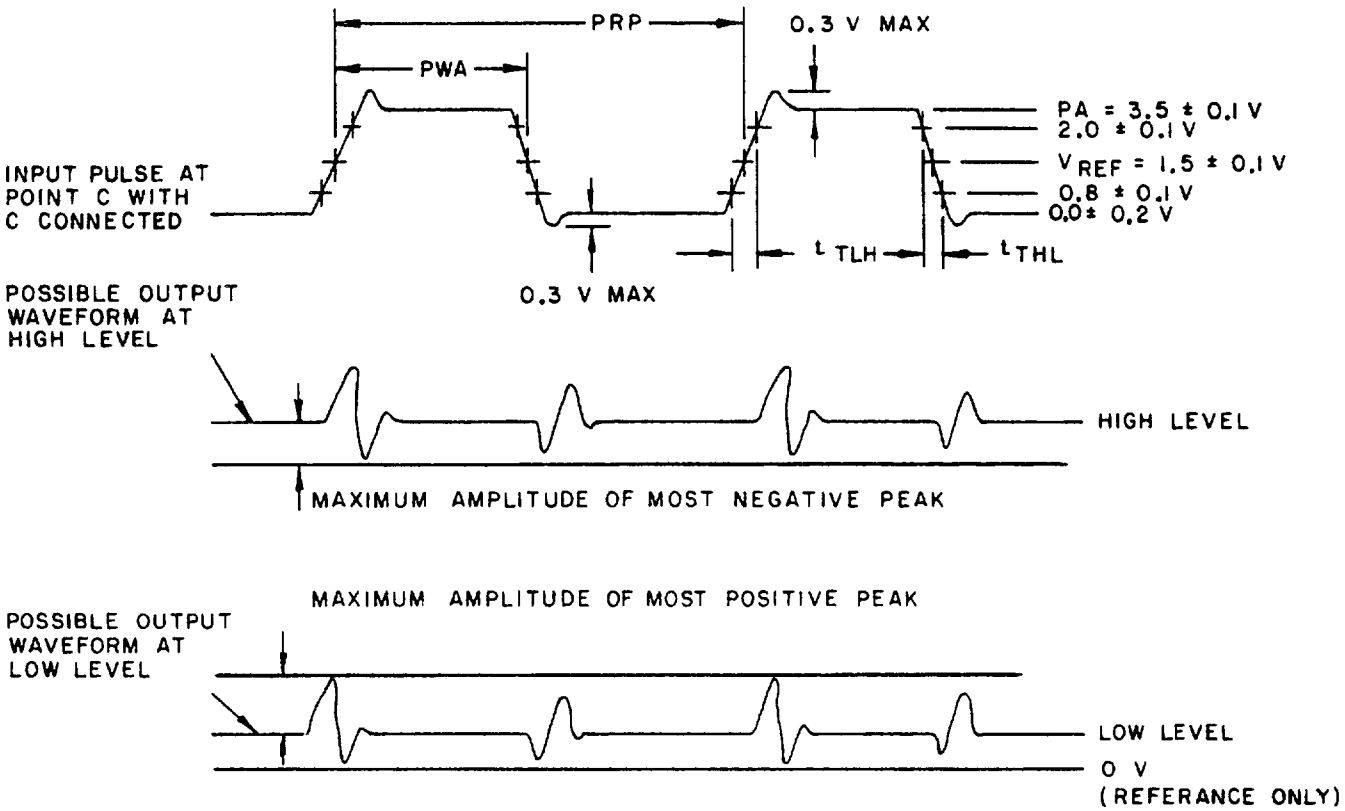
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NOTES.
1. R1 = 51Ω ±5%, 1/2 W.
2. Load outputs with load as specified on figure 12.

* FIGURE 18. Output crosstalk (low-level and high-level) test circuit for table XV.

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NOTES
1. Input pulse parameters.
   $t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between items Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of items Y01-1 and Y01-2 and test fixture and module ground pins shall be less than 1 inch (25.4 mm).

* FIGURE 19. Output crosstalk (low-level and high-level) test circuit waveforms for table XV.

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TABLE XVI. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
T	Power supply transient amplitude	0	400	mV	20,21

TABLE XVII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
U	Power supply filter capacitance	Open	1	1.6	3.0	µF	22

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TABLE XVIII. Input capacitance test.

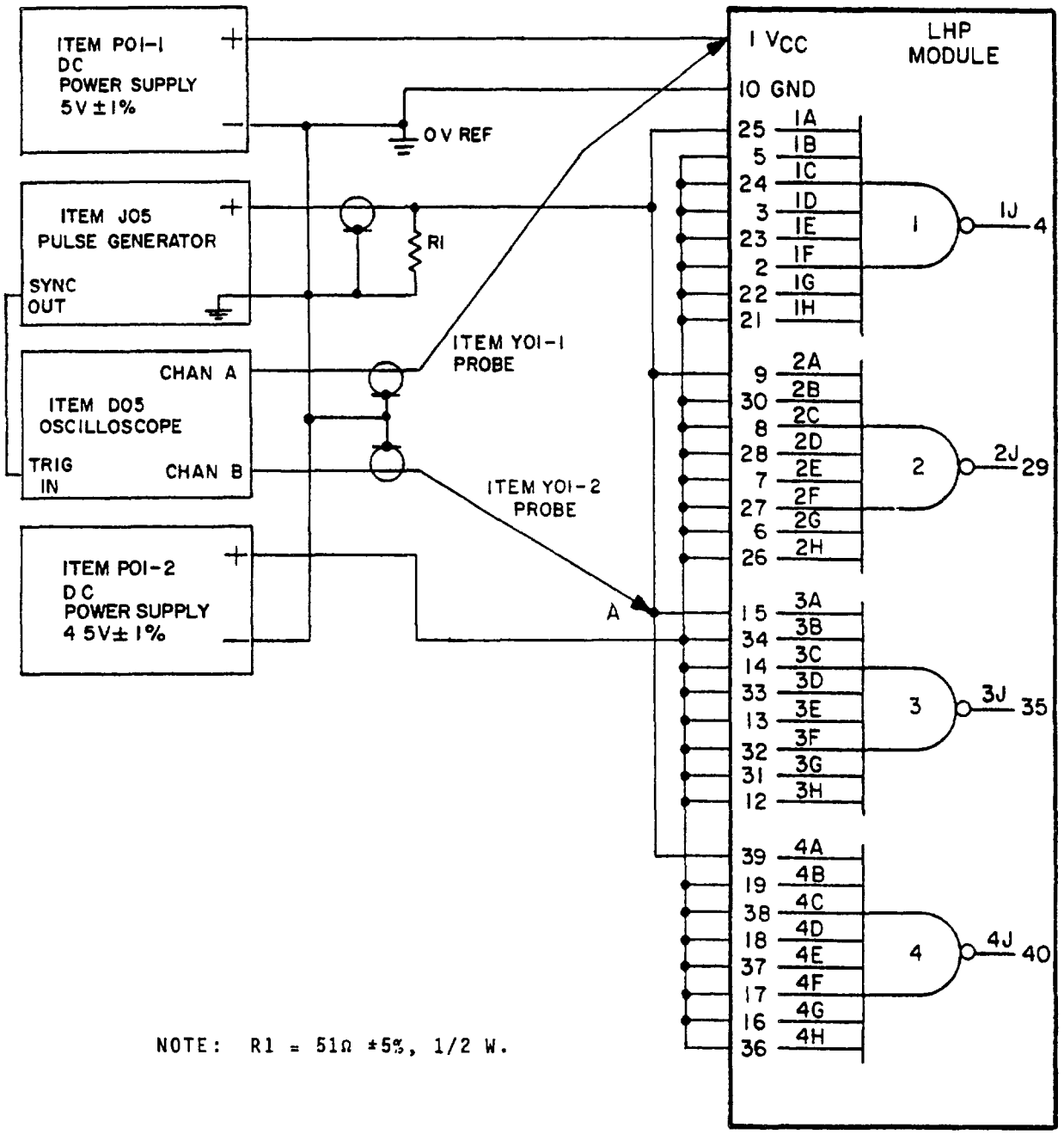
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V1	1A Input capacitance	25	Open	5	18	pF	22
V2	1B "	5	"	"	"	"	"
V3	1C "	24	"	"	"	"	"
V4	1D "	3	"	"	"	"	"
V5	1E "	23	"	"	"	"	"
V6	1F "	2	"	"	"	"	"
V7	1G "	22	"	"	"	"	"
V8	1H "	21	"	"	"	"	"
V9	2A "	9	"	"	"	"	"
V10	2B "	30	"	"	"	"	"
V11	2C "	8	"	"	"	"	"
V12	2D "	28	"	"	"	"	"
V13	2E "	7	"	"	"	"	"
V14	2F "	27	"	"	"	"	"
V15	2G "	6	"	"	"	"	"
V16	2H "	26	"	"	"	"	"
V17	3A "	15	"	"	"	"	"
V18	3B "	34	"	"	"	"	"
V19	3C "	14	"	"	"	"	"
V20	3D "	33	"	"	"	"	"
V21	3E "	13	"	"	"	"	"

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TABLE XVIII. Input capacitance test - Continued.

Test condition	-	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
V22	3F	Input capacitance	32	Open	5	18	pF	22
V23	3G	"	31	"	"	"	"	"
V24	3H	"	12	"	"	"	"	"
V25	4A	"	39	"	"	"	"	"
V26	4B	"	19	"	"	"	"	"
V27	4C	"	38	"	"	"	"	"
V28	4D	"	18	"	"	"	"	"
V29	4E	"	37	"	"	"	"	"
V30	4F	"	17	"	"	"	"	"
V31	4G	"	16	"	"	"	"	"
V32	4H	"	36	"	"	"	"	"

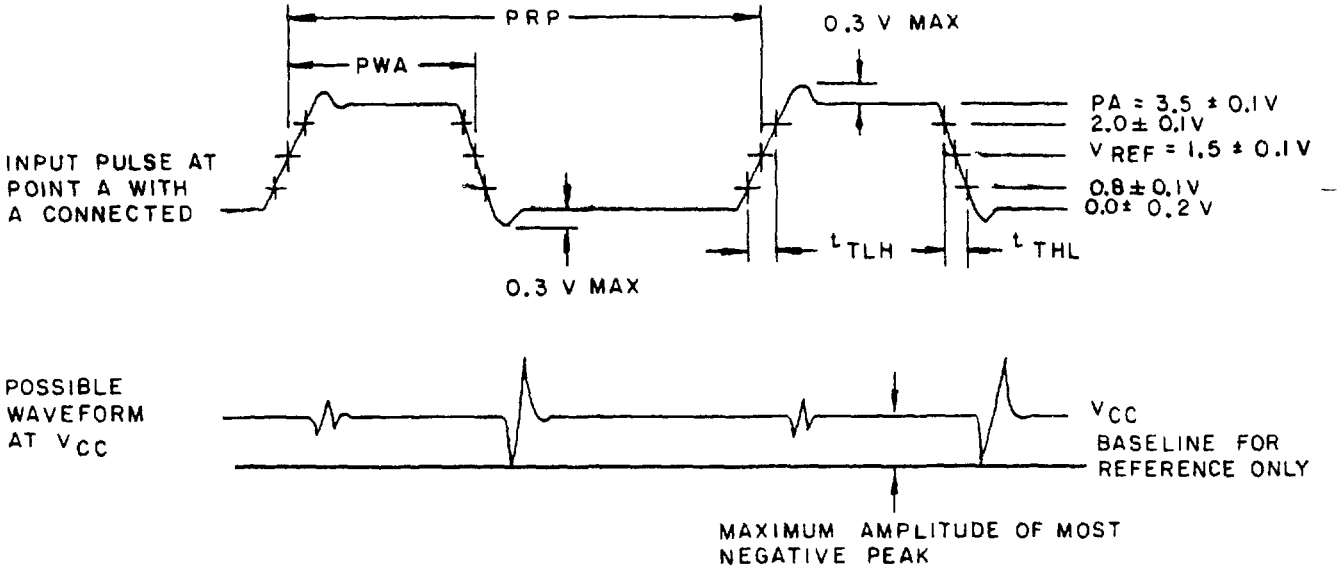
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 20. Power supply transient amplitude test circuit for table XVI.

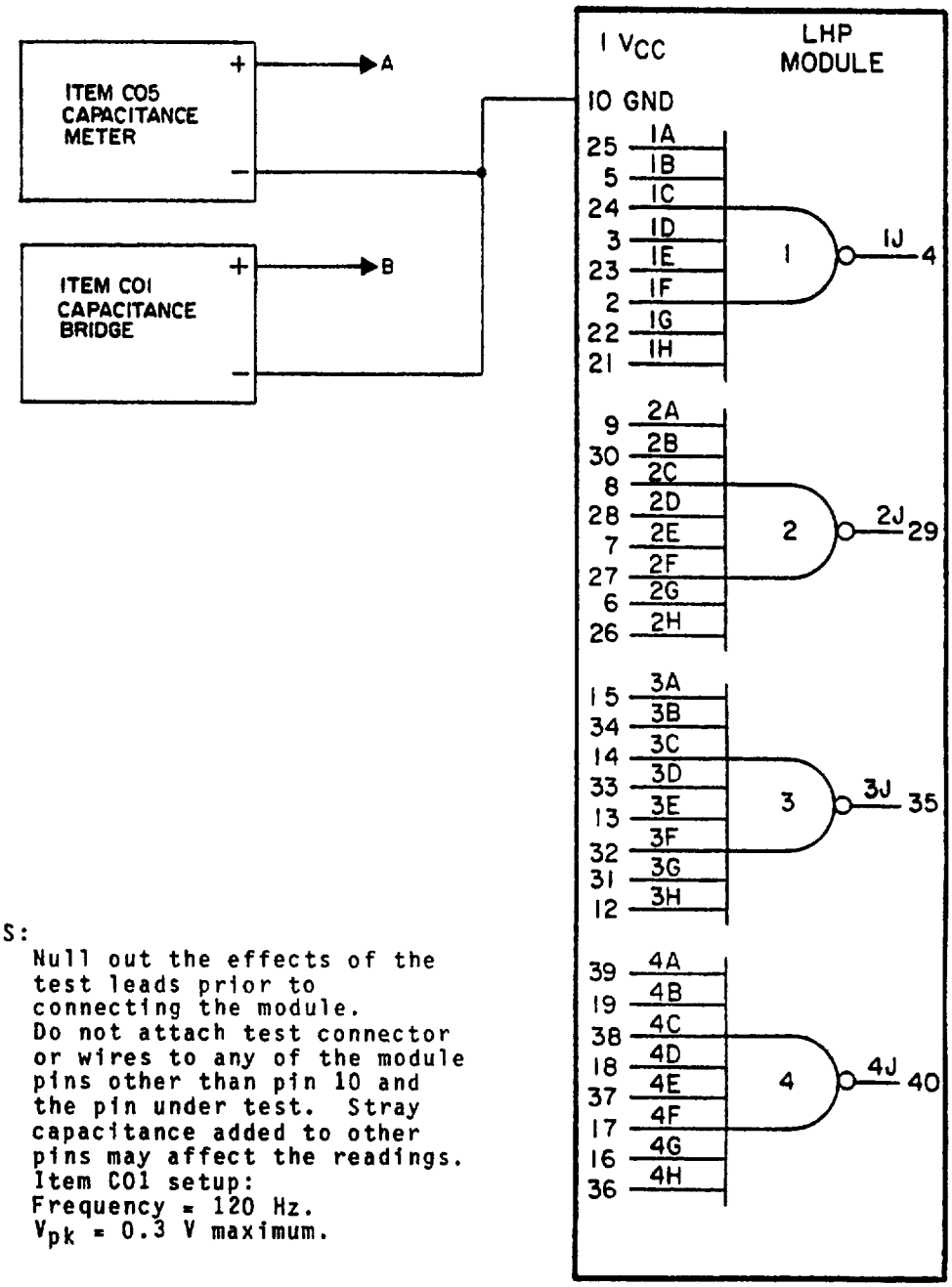
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NOTE. Input pulse parameters:
$t_{TLH} = t_{THL} = 3.5 \pm 1.5$ ns (verified at each input pin).
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns.

FIGURE 21. Power supply transient amplitude test circuit waveforms for table XVI.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item CO1 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

* FIGURE 22. Power supply filter capacitance and input capacitance test circuit for tablex XVII and XVIII.

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